

DATA ANALYTICS & BUSINESS INTELLIGENCE PROJECT

HOTEL BOOKING ANALYSIS

Revenue · Marketing · Cancellations · Customer Behavior

PRESENTED BY

Karima Saad · Nourhan Wael · Elsayed Elsayaf
Walid Nofal · Hana Mohamed · Eman Elnahas

KARIMA SAAD

Project Overview · Data Source · Python Exploration + Cleaning



From raw hotel bookings to an analysis-ready dataset

Project Overview



This project analyzes ~117,000 bookings across City Hotel and Resort Hotel (2015–2017) to surface revenue, marketing, cancellation, and customer behavior patterns.

117K

BOOKING RECORDS

2015...2017

TIME WINDOW

2

HOTEL PROPERTIES

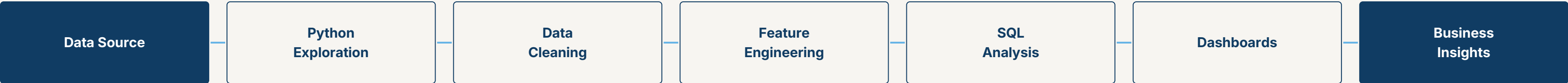
36

FIELDS PER BOOKING

ANALYSIS PILLARS

Revenue Analysis · Booking Analysis · Cancellation Analysis · Marketing Analysis · Customer Behavior

END-TO-END PIPELINE



Data Source & Dataset



Two CSV artifacts anchor the project: the raw hotel_booking.csv and the cleaned hotel_bookings_cleaned.csv. Together they hold ~117,000 records with 36 fields each, grouped below by analytic domain.

Stay Details • lead_time • arrival_date • weekend_nights • week_nights • room_type	Guest Details • adults • children • babies • country • repeat_guest	Booking / Marketing • market_segment • distribution_channel • deposit_type • booking_changes	Financial • ADR • agent • company • special_requests	Outcome • is_canceled • reservation_status_date
--	---	--	--	---

Data Exploration



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('hotel_booking.csv')
df.head()
df.tail()
df.shape          # (119390, 36)
df.info()
df.describe()
```

read_csv	Load the raw booking CSV into a DataFrame.
head / tail	Inspect first and last rows for structure.
shape	Confirm record and column counts.
info	Data types and non-null counts per column.
describe	Summary statistics for numeric fields.

Data Cleaning

```
df.duplicated().sum()
df.isnull().sum()
df.drop(columns=['company'], inplace=True)
df['agent'] = df['agent'].fillna(0).astype(int)
df['country'] = df['country'].fillna('Unknown')
df['children'] = df['children'].fillna(0).astype(int)
df.drop(columns=['name', 'email',
                 'phone-number', 'credit_card'],
        inplace=True)
df['reservation_status_date'] = pd.to_datetime(
    df['reservation_status_date'])
df = df[(df['adults']+df['children']+df['babies'])>0]
df = df[(df['adr']>0) & (df['adr']<5000)]
df.reset_index(drop=True, inplace=True)
```

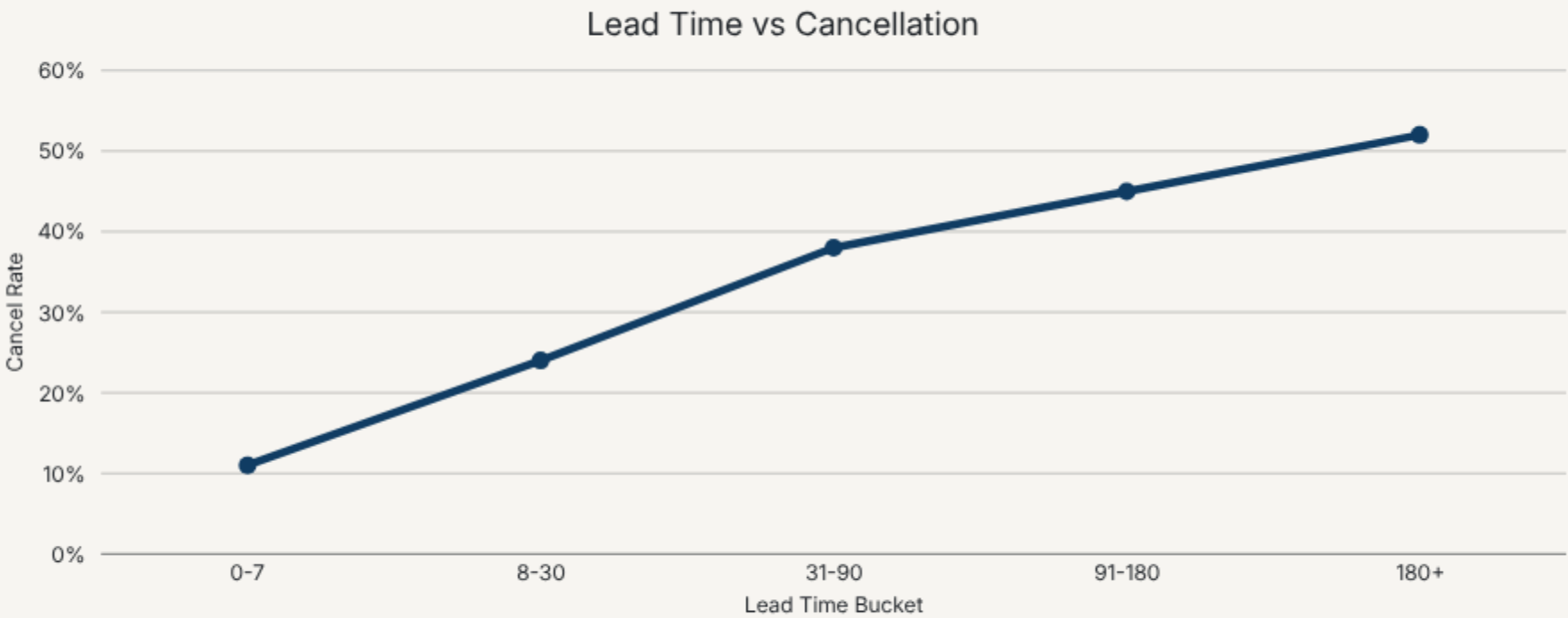
- 01 Check duplicates & missing values
- 02 Drop 'company' (excessive nulls)
- 03 Fill 'agent' with 0
- 04 Fill 'country' with 'Unknown'
- 05 Fill 'children' with 0
- 06 Drop PII: name, email, phone, credit_card
- 07 Cast 'children' & 'agent' to int
- 08 Parse 'reservation_status_date' → datetime
- 09 Remove zero-guest rows
- 10 Filter ADR ∈ (0, 5000)
- 11 Reset index

Feature Engineering & EDA Visualizations

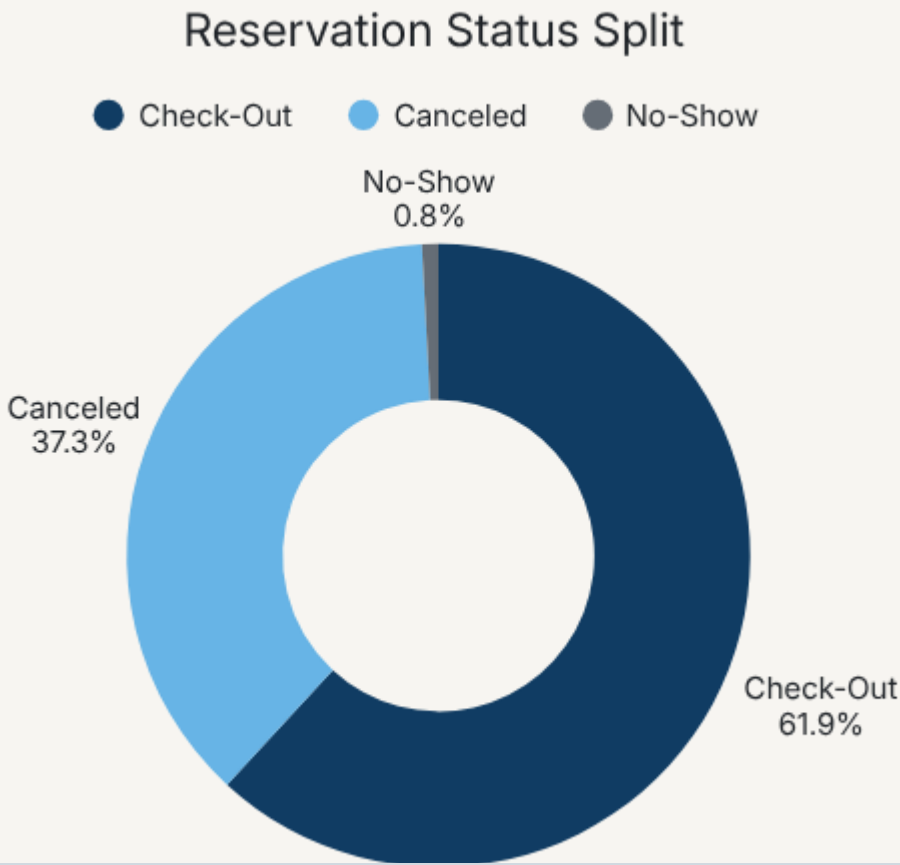
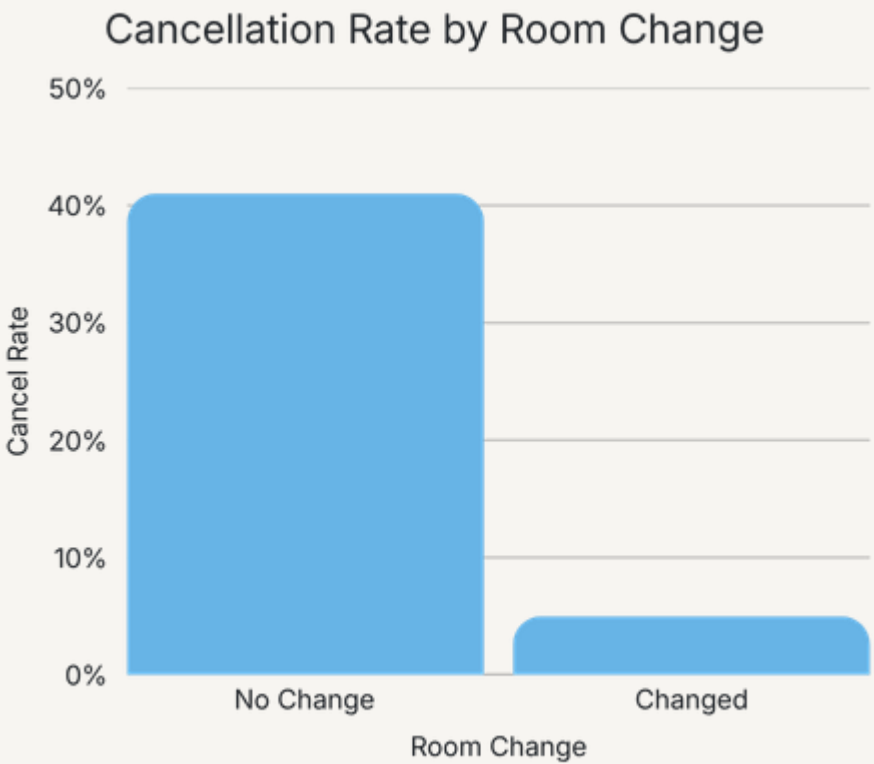


NEW FEATURES

<code>total_stay</code>	<code>weekend_nights + week_nights</code>
<code>total_guests</code>	<code>adults + children + babies</code>
<code>reservation_year</code>	<code>extracted from reservation_status_date</code>
<code>reservation_month</code>	<code>extracted from reservation_status_date</code>
<code>room_type_changed</code>	<code>reserved != assigned room type</code>



Longer lead times correlate with higher cancellation risk.



Cancellations run ~37.5% of all bookings.



HOTEL BOOKING
— ANALYSIS —

02 /
06

NOURHAN WAEL

SQL Query 1 • Revenue Analysis + Dashboard

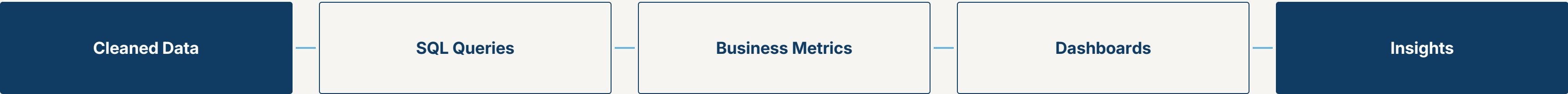


From cleaned bookings to
realized and lost revenue

SQL Analysis Workflow



SQL complements Python: cleaned data is loaded into a SQL environment where aggregations, window functions, and joins power KPIs that feed the dashboards.



```
-- 1_Revenue Query.sql
SELECT hotel,
       SUM((stays_in_weekend_nights + stays_in_week_nights) * adr)
       AS total_revenue
FROM hotel_bookings_cleaned
WHERE is_canceled = 0
GROUP BY hotel
ORDER BY total_revenue DESC;
```

QUERY 1 — REVENUE ANALYSIS

Establishes hotel-level revenue baselines, ADR seasonality, and lost revenue impact.

Revenue — Business Questions



Q1	Total revenue by hotel
Q2	Average ADR by month
Q3	Lost revenue by month (cancellations)
Q4	Average ADR by market segment
Q5	Hotel revenue and average lead time
Q6	Revenue and booking count by meal plan
Q7	Top 5 countries by revenue

```
-- Sql/1_Revenue Query.sql
-- Total revenue by hotel
SELECT hotel, SUM(...) AS total_revenue ...

-- Average ADR by month
SELECT arrival_date_month, AVG(adr) ...

-- Lost revenue by month
SELECT arrival_date_month,
       SUM((weekend+week)*adr) AS lost_revenue
WHERE is_canceled = 1 ...

-- Top 5 countries by revenue
SELECT TOP 5 country, SUM(...) ...
GROUP BY country ORDER BY revenue DESC;
```

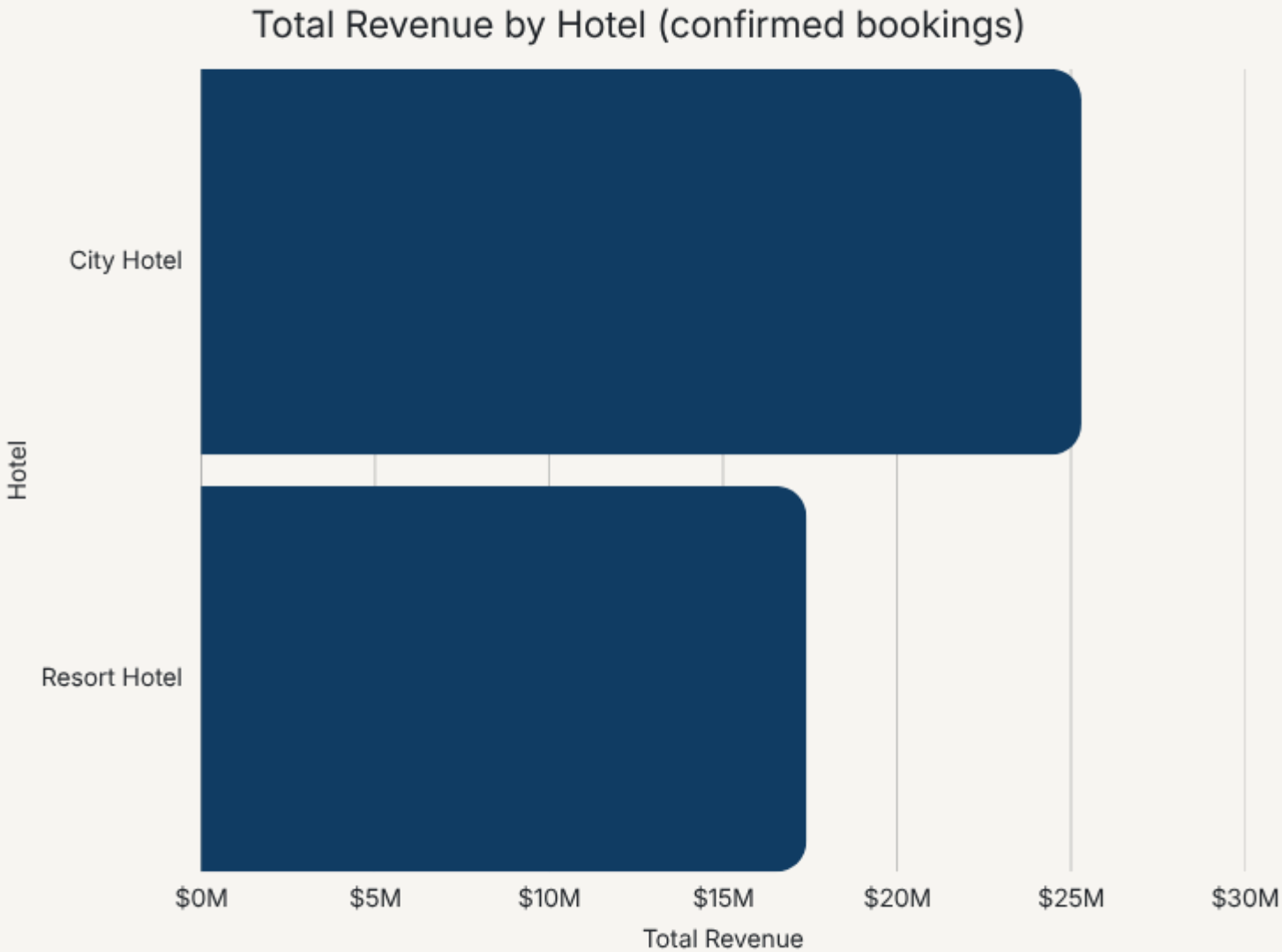
Total Revenue by Hotel

```
SELECT hotel,
  SUM((stays_in_weekend_nights + stays_in_week_nights) * adr)
    AS Total_Revenue
FROM hotel_bookings_cleaned
WHERE is_canceled = 0
GROUP BY hotel
ORDER BY Total_Revenue DESC;
FORMULA
Total Revenue = (weekend_nights + week_nights) × ADR · filtered to is_canceled = 0
```

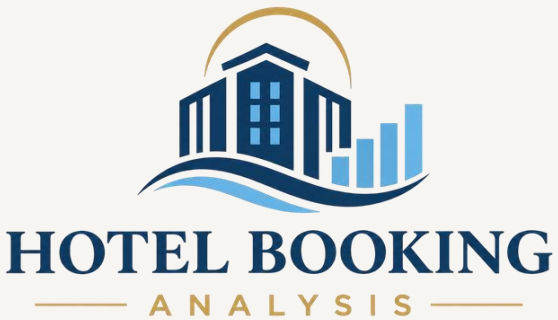
KEY OBSERVATION

City Hotel outperforms Resort Hotel on realized revenue, driven by higher booking volume despite comparable ADR.

	hotel	Total_Revenue
1	City Hotel	14385341
2	Resort Hotel	11601634



ADR & Lost Revenue by Month



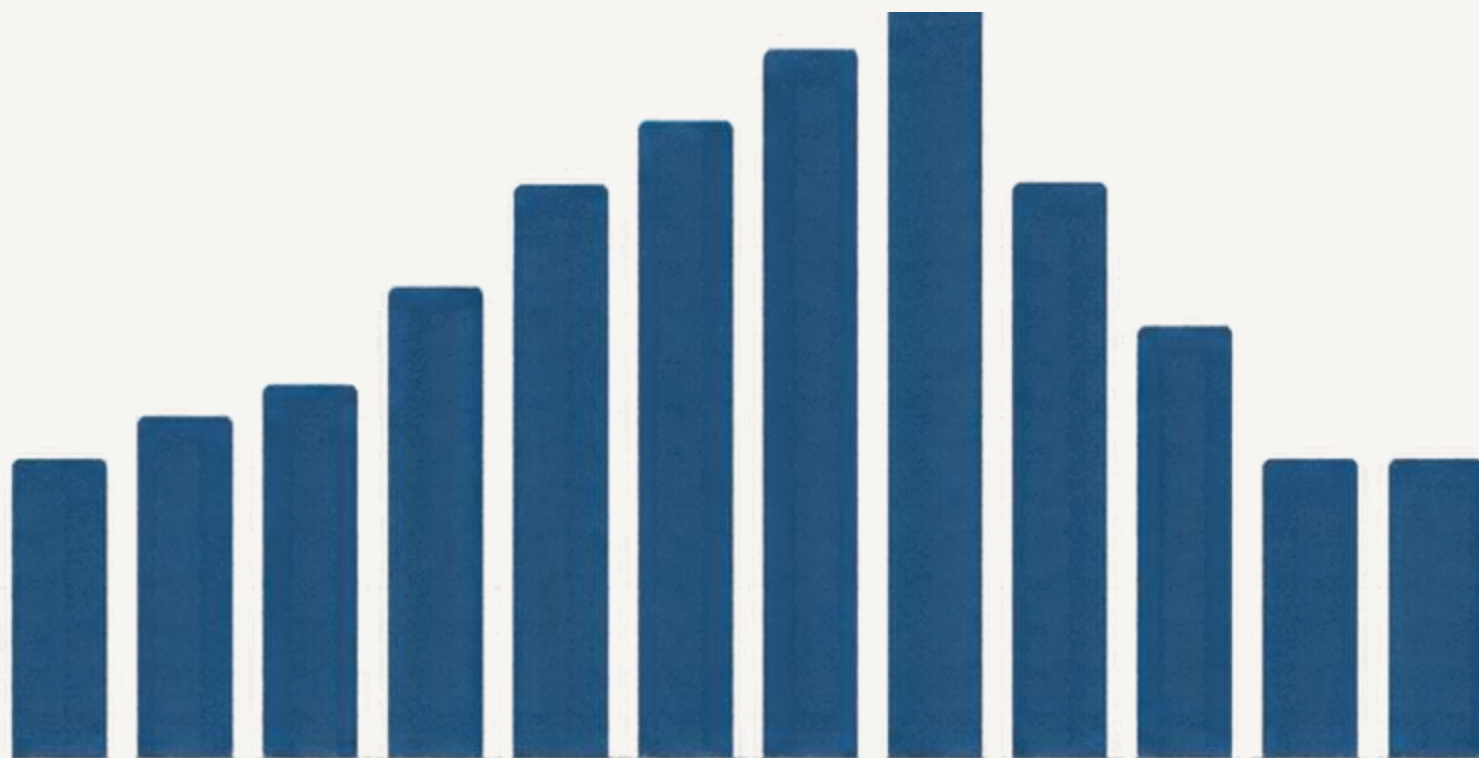
```
-- Avg ADR by arrival month (confirmed)
SELECT arrival_date_month, AVG(adr) AS avg_adr
WHERE is_canceled = 0 GROUP BY arrival_date_month;
-- Lost revenue by month (canceled)
SELECT arrival_date_month,
       SUM((weekend+week)*adr) AS lost_revenue
WHERE is_canceled = 1 GROUP BY arrival_date_month;
```

Together these metrics separate captured revenue from missed opportunity — a lens for pricing strategy and cancellation mitigation.

Realized — Average ADR by Month



Lost — Revenue from Cancellations by Month



```
Q4 — Average ADR by Market Segment
SELECT market segment, Avg(adj) AS Average_ADR
WHERE is(cancelled = 0
GROUP BY market segment
ORDER BY Average ADR DESC,
```



Q4 — Average ADR by Market Segment

This query compares the average daily rate across market segments for confirmed bookings, highlighting which segments command higher ADR and informing targeted pricing and segmentation strategies.

Revenue Analysis • Q5–Q7





Q5. Hotel Revenue Plus Average Lead Time

Using confirmed bookings only, calculate total revenue and average lead time (days) by hotel.

Returns:

- ✓ Hotel
- ✓ Average_Lead_Time_Days
- ✓ Total_Revenue

```
SELECT HOTEL,
_AVG(AVERAGE_LEAD_TIME_DAYS) AS
AVERAGE_LEAD_TIME_DAYS, SUM(ADR * (1 - DISCOUNT))
AS TOTALREVENUE, COUNT (*) AS NUMBER_OF_BOOKINGS
FROM BOOKINGS
WHERE IS CANCELED = 0 AND
GROUP BY HOTEL
ORDER BY TOTAL_REVENUE DESC,
```



Q6. Revenue and Booking Count by Meal Plan

Using confirmed bookings only, calculate total revenue and number of bookings by meal plan.

Returns:

- ✓ Meal
- ✓ Total_Revenue
- ✓ Number_Of_Bookings

```
SELECT MEAL,
SUM(ADR * (1 - DISCOUNT)) AS TOTALREVENUE,
COUNT (*) AS NUMBER_OF_BOOKINGS
FROM BOOKINGS
WHERE IS CANCELED = 0 AND
GROUP BY MEAL
ORDER BY TOTAL_REVENUE DESC;
```



Q7. Top 5 Countries by Revenue

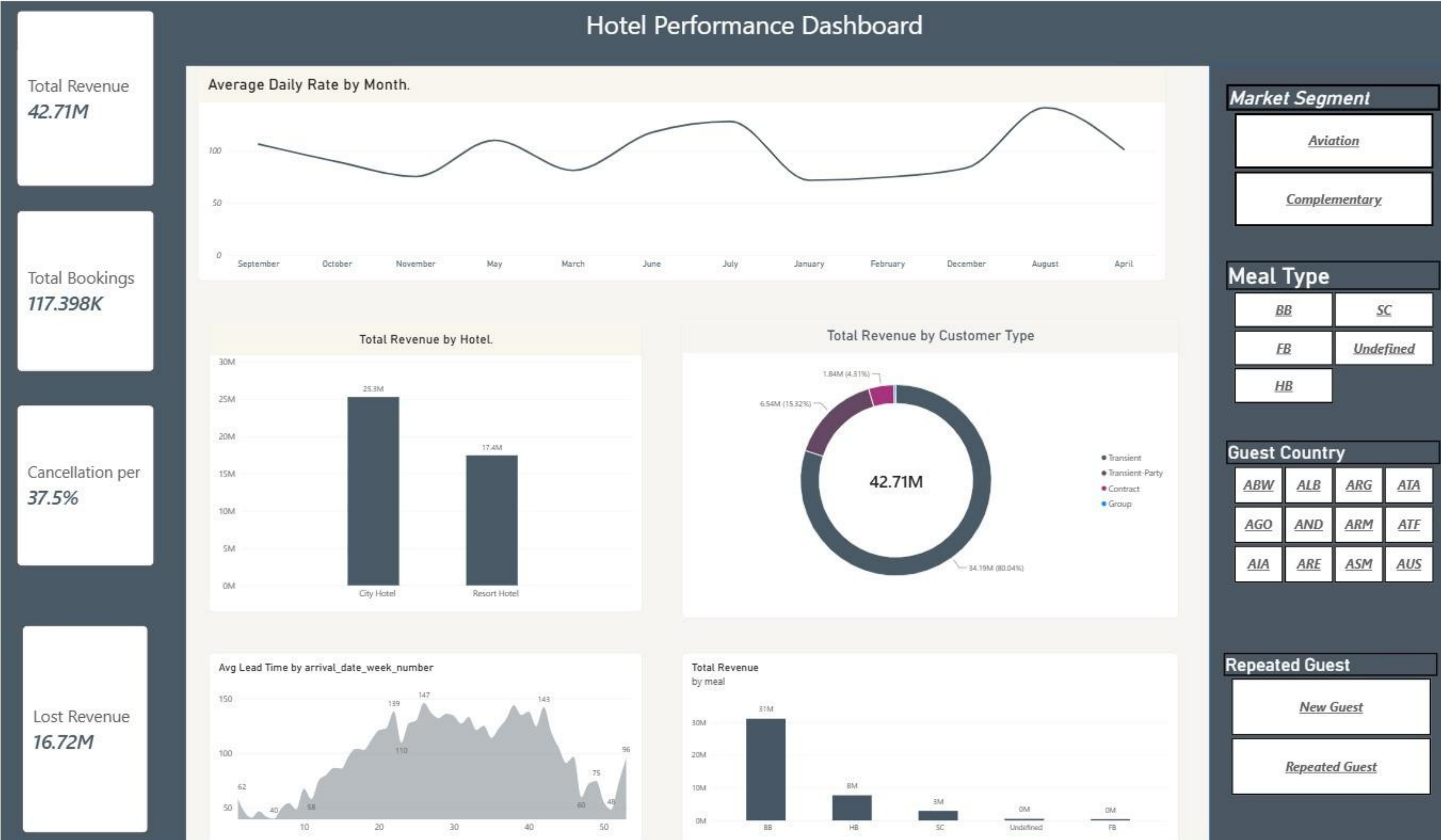
Using confirmed bookings only, exclude null countries and return the top 5 countries by total revenue.

Returns:

- ✓ Country
- ✓ Total_Revenue

```
SELECT COUNTRY,
SUM(ADR * (1 - DISCOUNT)) AS TOTALREVENUE,
FROM BOOKINGS
WHERE IS CANCELED = 0 AND
AND COUNTRY IS NOT NULLL
GROUP BY COUNTRY
ORDER BY TOTAL_REVENUE DESC
LIMIT 5;
```

Revenue / Hotel Performance Dashboard



Revenue — Key Findings



OBSERVED DATA

- City Hotel generates higher realized revenue than Resort Hotel.
- Cancellations account for ~\$16.72M in lost revenue across the dataset.
- ADR varies materially by month and by market segment.

BUSINESS INTERPRETATION

- Reducing cancellation rate is the largest single lever to recover revenue.
- Seasonal ADR peaks (Jul–Aug) support demand-based pricing strategy.
- Segment-level ADR gaps inform channel-specific commercial policies.



HOTEL BOOKING
— ANALYSIS —

03 / 06

ELSAYED ELSAWAF

SQL Query 2 • Market Analysis

From market segments to distribution performance



Market Analysis — Scope



- 1

Top 10 countries by revenue
- 2

Revenue by market segment
- 3

Market segment booking share
- 4

Cancellation rate by distribution channel
- 5

Top 10 agents by confirmed bookings
- 6

Top 10 agents by revenue
- 7

Repeat guests by market segment
- 8

Average lead time by market segment

```
-- Sql/2_Market Analysis.sql
-- Revenue by country
SELECT TOP 10 country,
       SUM((weekend+week)*adr) AS revenue
FROM hotel_bookings_cleaned
WHERE is_canceled = 0
GROUP BY country ORDER BY revenue DESC;

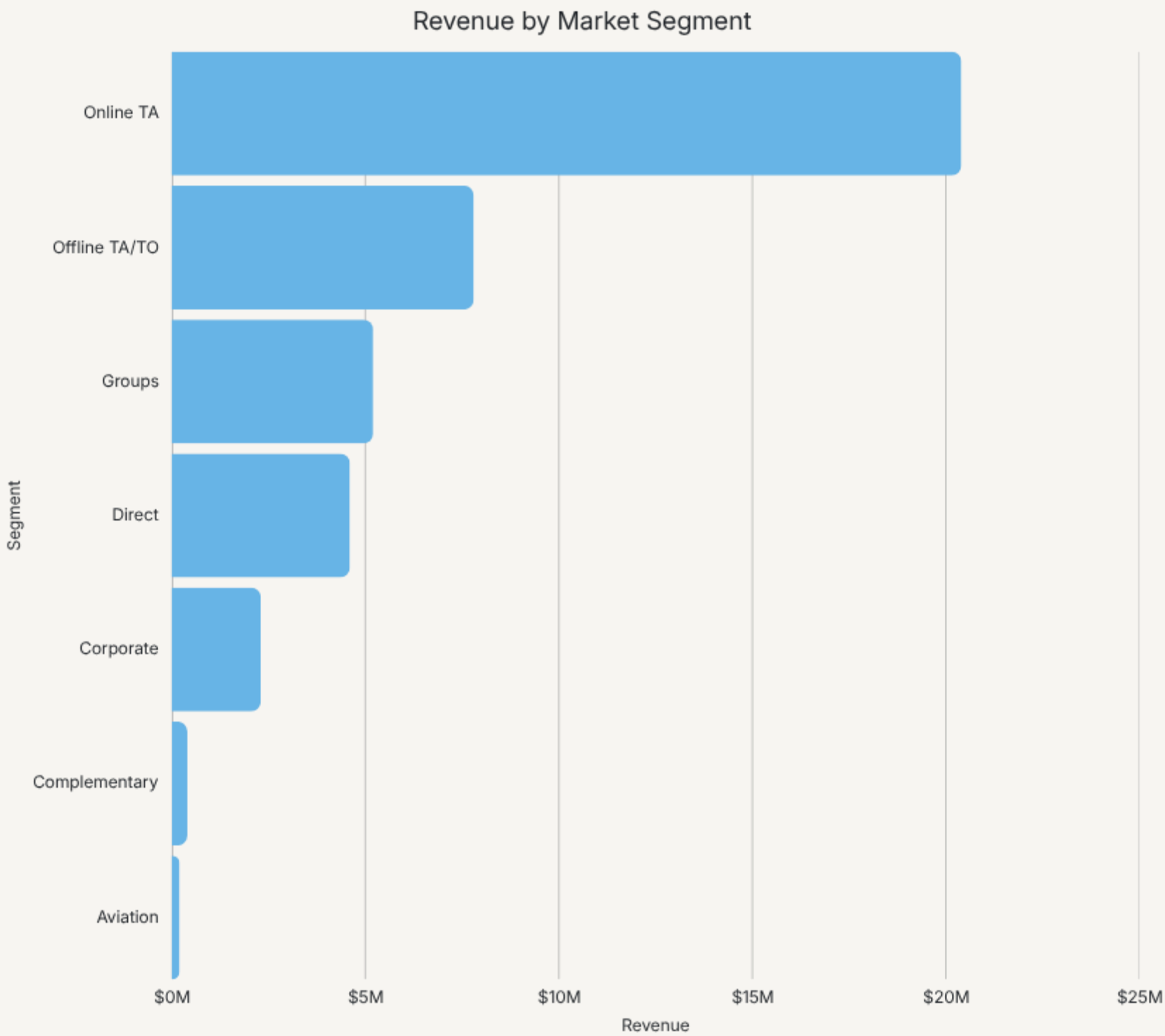
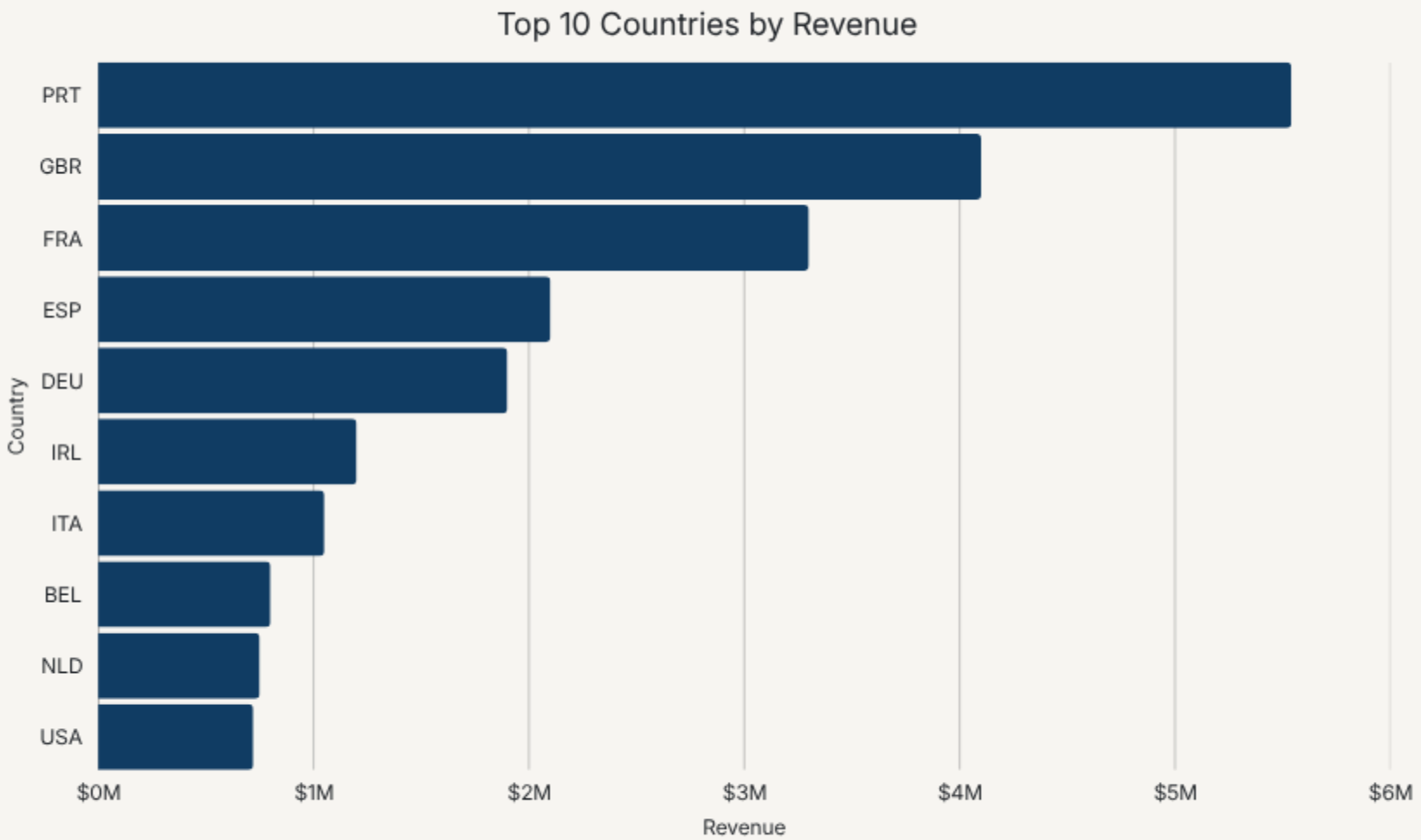
-- Market segment booking share
SELECT market_segment,
       COUNT(*) * 100.0 / SUM(COUNT(*)) OVER ()
       AS booking_share_pct
FROM hotel_bookings_cleaned
GROUP BY market_segment;
```

SQL Query 2 · Questions 1

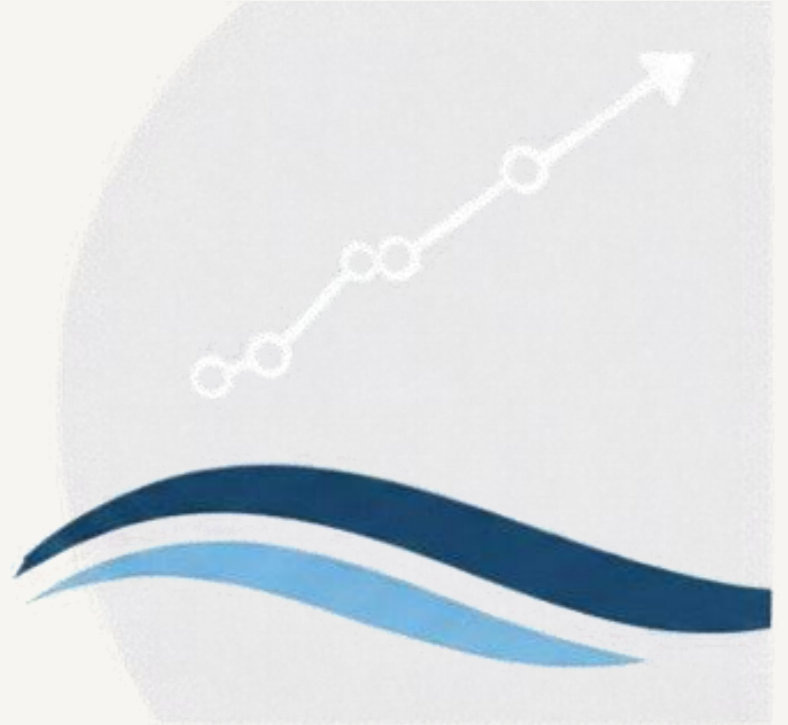
Top Countries & Market Segments

```
SELECT TOP 10 country,
COUNT(*) AS bookings,
SUM((weekend+week)*adr) AS revenue
FROM hotel_bookings_cleaned
WHERE is_canceled = 0
GROUP BY country ORDER BY revenue DESC;
```

Portugal leads country revenue at ~\$5.54M — a home-market advantage that dwarfs the second-tier European feeder markets.



SQL QUERY 2 • QUESTIONS 2



Top countries and market segments

02 Revenue by Market Segment

OUTPUT COLUMNS

market_segmenttotal_bookingstotal_revenueavg_adj

SQL LOGIC

```
SELECT
market_segment,
COUNT(*) AS total_bookings,
ROUND(SUM( adr * (stays_in_weekend_nights + stays_in_week_night
AS total_revenue,
ROUND(AVG(adr), 2) AS avg_adr
FROM dbo.[hotel_bookings_cleaned.csv]
WHERE is_canceled = 0
GROUP BY market_segment
ORDER BY total_revenue DESC;
```

market_segment	total_bookings	total_revenue	avg_adr
Online TA	35370	13707139.09	115.01
Offline TA/TO	15614	5659291.94	84.98
Direct	10458	4097842.4	116.17
Groups	7484	1869129.52	79.5
Corporate	4224	577892.19	68.36
Aviation	180	70868.36	102.25
Complementary	58	4812.53	34.29

SQL Query 2 • Questions 3

Booking Share Percentage — Explained

```
SELECT market_segment,  
       CAST(  
         COUNT(*) * 100.0 / SUM(COUNT(*)) OVER ()  
         AS DECIMAL(5,2)) AS booking_share_pct  
FROM hotel_bookings_cleaned  
GROUP BY market_segment;
```

COUNT(*)
Counts bookings within each market_segment group.

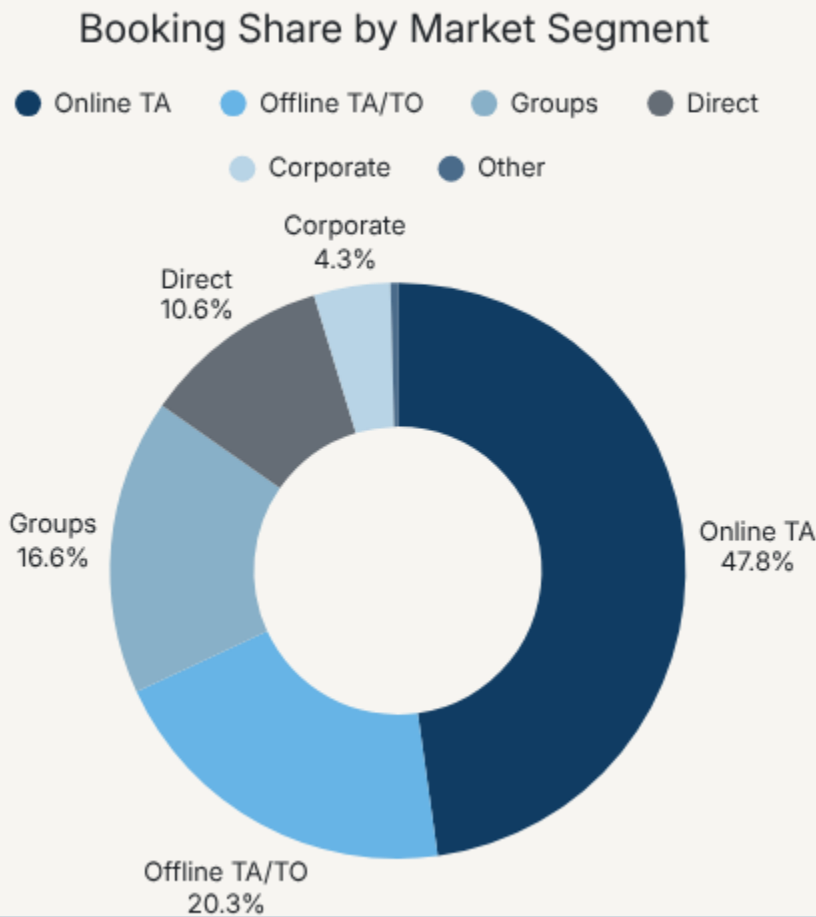
IMPORTANT DISTINCTION

Booking share is the proportion of BOOKINGS per segment — not revenue share. A segment can hold a large booking share while contributing lower revenue if its ADR is below average.

SUM(COUNT(*)) OVER ()
Window sum: total bookings across ALL segments.

Division
Segment count ÷ overall total yields the proportion.

CAST as DECIMAL(5,2)
Formats the result as a clean percentage.



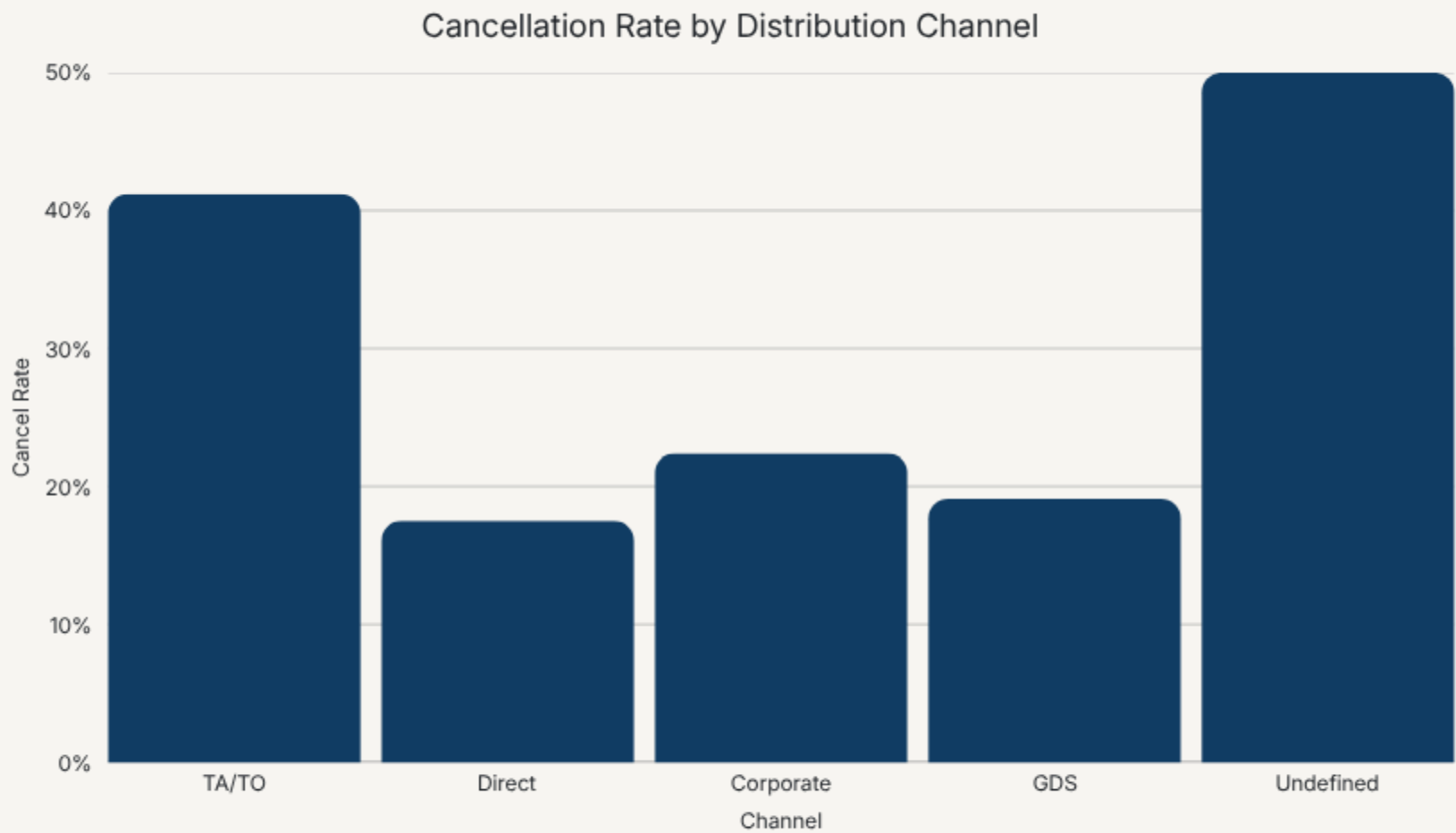
	market_segment	total_bookings	booking_share_pct
1	Online TA	56089	47.78
2	Offline TA/TO	23884	20.34
3	Groups	19557	16.66
4	Direct	12361	10.53
5	Corporate	5211	4.44
6	Aviation	231	0.20
7	Complementary	63	0.05
8	Undefined	2	0.00

SQL Query 2 · Questions 4

Distribution Channel & Agent Analysis

```
SELECT distribution_channel,
       COUNT(*) AS total_bookings,
       SUM(CASE WHEN is_canceled=1 THEN 1 ELSE 0 END)
         AS canceled_bookings,
       100.0 * SUM(CASE WHEN is_canceled=1 THEN 1 ELSE 0 END)
         / COUNT(*) AS cancellation_rate_pct
GROUP BY distribution_channel;
```

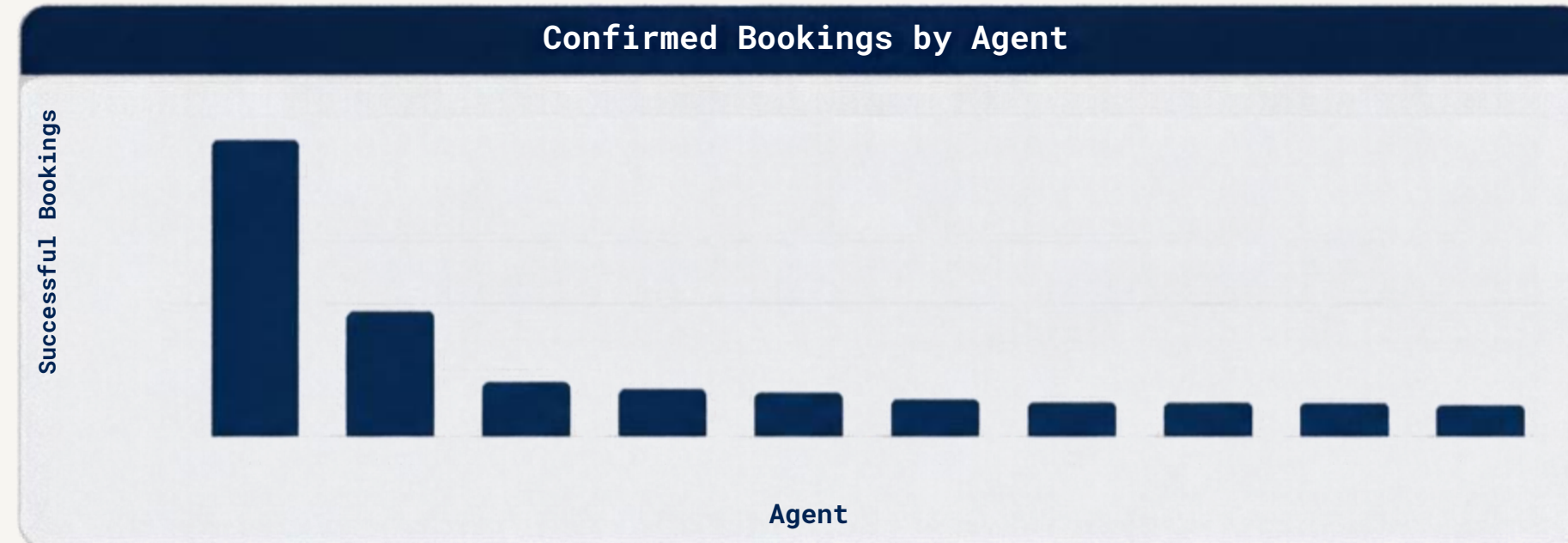
	distribution_channel	total_bookings	canceled_bookings	cancellation_rate_pct
1	Undefined	5	4	80.00
2	TATO	96888	40064	41.35
3	Corporate	6483	1448	22.34
4	GDS	189	37	19.58
5	Direct	13833	2457	17.76



SQL Query 2 • Questions 5

Top 10 Agents by Confirmed Bookings

```
SELECT TOP 10
  agent,
  COUNT(*) AS successful_bookings,
  ROUND(SUM(adr * (stays_in_weekend_nights
    + stays_in_week_nights)), 2) AS total_revenue,
  ROUND(AVG(adr), 2) AS avg_adr
FROM dbo.[hotel_bookings_cleaned.csv]
WHERE is_canceled = 0
  AND agent <> 0
GROUP BY agent
ORDER BY successful_bookings DESC;
```



Confirmed Bookings		
Rank	Agent	successful_bookings
1	9	23000
2	240	9200
3	14	3400
4	7	2800
5	6	2400
6	3	1800
7	1	1600
8	40	1500
9	250	1400
10	241	1300

Note: Confirmed bookings only; agent = 0 excluded.

SQL QUERY 2 • QUESTION 6

Calculate each market segment's percentage share of total bookings (not revenue share).



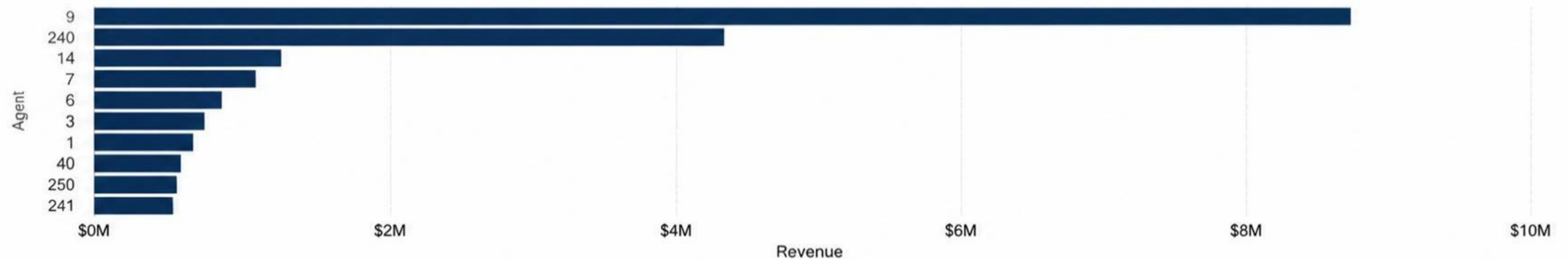
SQL QUERY

```
SELECT TOP  
AGENT,  
COUNT(*) AS SUCCESSFUL BOOKINGS,  
ROUND(SUM ADR * (STAYS_IN_WEEKNIGHT_NIGHTS + STAYS_IN_WEEK_NIGHTS), 2)  
AS TOTAL_REVENUE,  
ROUND(AVG ADR) AS AVG_ADC  
FROM DBO.[HOTEL_BOOKINGS_CLAANED.CSV]  
WHERE IS_CANCELED = 0  
AND AGENT <= 250  
GROUP BY AGENT  
ORDER BY TOTAL_REVENUE DESC;
```

TOP 10 AGENTS BY REVENUE

	agent	successful_bookings	total_revenue	avgadr
1	9	18590	6990224.55	123.46
2	240	8308	3596513.26	110.71
3	250	2344	1406148.9	130.78
4	14	2967	1161724.24	126.1
6	7	9943	978087.77	94.95
6	40	938	657565.64	68.68
7	241	1469	640851.19	98.17
8	6	2190	510401.09	92.18
9	28	1527	420434.74	80.83
10	8	1070	414317.71	108.82

TOP 10 AGENTS BY REVENUE



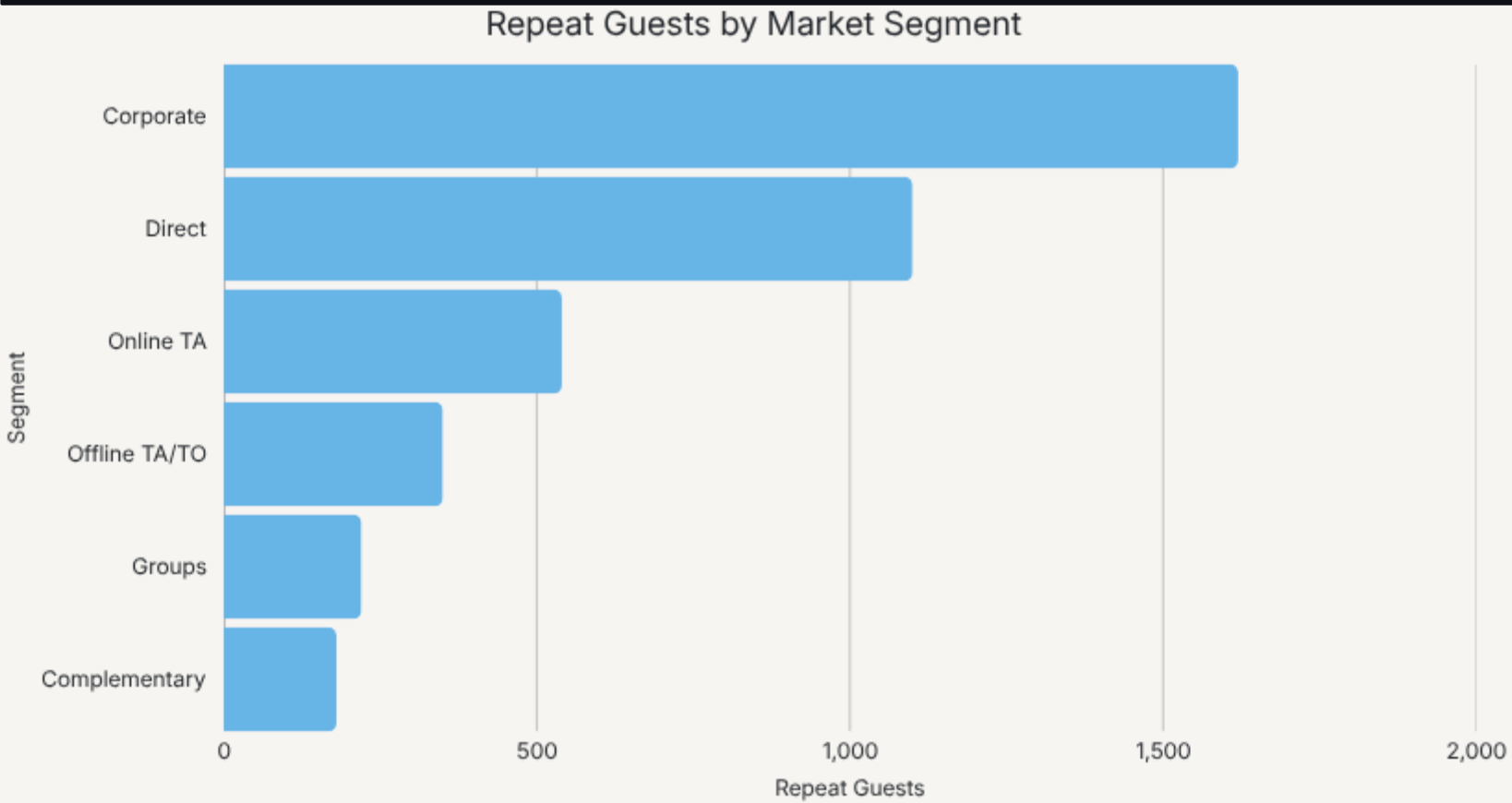
SQL QUERY 2 • QUESTION 7

7 Repeat Guests by Market Segment

```
SELECT market_segment,
       COUNT(*) AS repeat_guest_bookings
FROM hotel_bookings_cleaned
WHERE is_repeated_guest = 1
GROUP BY market_segment;

SELECT market_segment,
       ROUND(CAST(AVG(lead_time) AS FLOAT),1) AS avg_lead_time
GROUP BY market_segment;
```

	market_segment	repeated_guests	repeated_guest_revenue	avg_ad
1	Corporate	1350	139728.64	64.35
2	Direct	653	186513.96	89.21
3	Online TA	441	89500.59	84.74
4	Offline TA/TO	205	44070.64	64.72
5	Aviation	50	15562	104.08
6	Groups	38	8813.57	93.36
7	Complementary	5	420	48.68



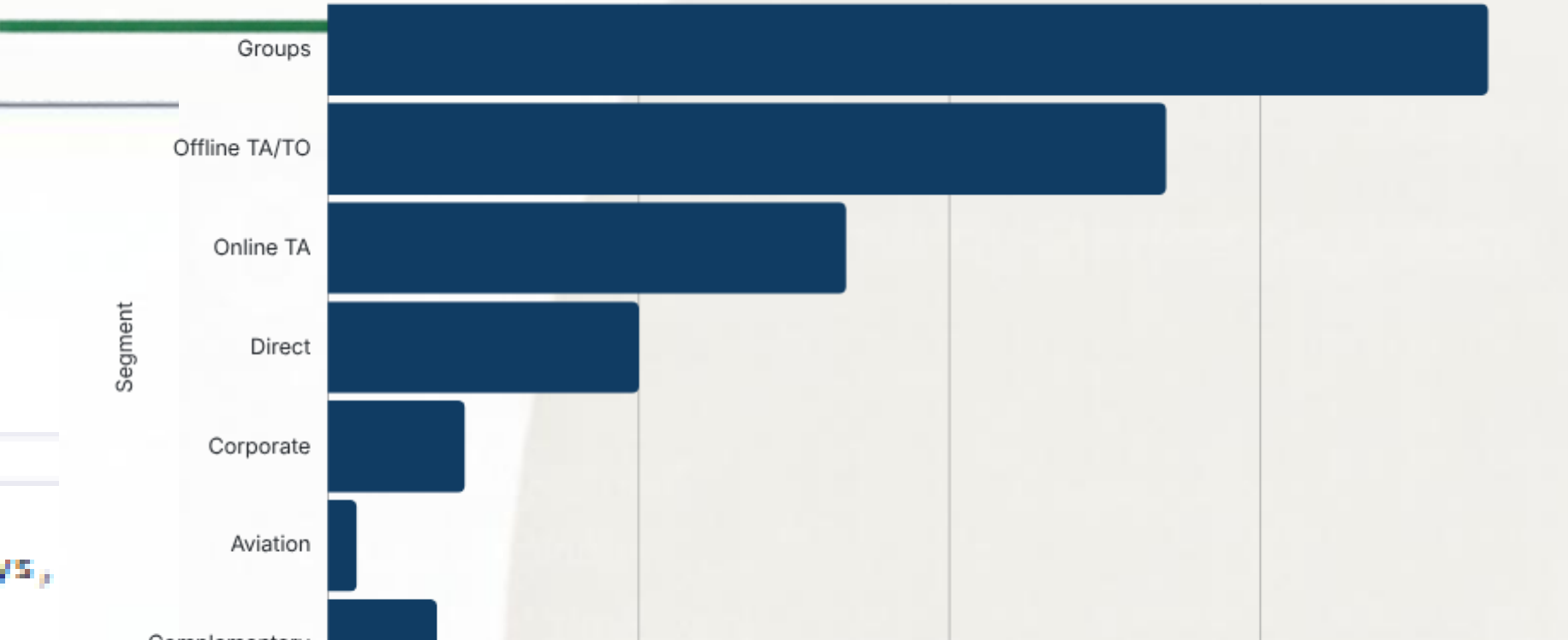
SQL QUERY 2 • QUESTIONS 8

Average Lead Time by Market Segment

8

Average Lead Time by Market Segment

```
SELECT
    market_segment,
    COUNT(*) AS total_bookings,
    ROUND(AVG(CAST(lead_time AS FLOAT)),2) AS avg_lead_time_days,
    ROUND(AVG(CAST(adr AS FLOAT)),2) AS avg_ad
FROM dbo.[hotel_bookings_cleaned.csv]
GROUP BY market_segment
ORDER BY avg_lead_time_days DESC;
```



	market_segment	total_bookings	avg_lead_time_days	avg_ad
1	Groups	19557	187.94	80.51
2	Offline TA/TO	23884	135.96	88.34
3	Online TA	56089	83.36	117.99
4	Direct	12361	50.16	117.7
5	Complementary	63	37.75	34.04
6	Corporate	5211	22.29	70.48
7	Aviation	231	4.54	102.74
8	Undefined	2	1.5	15

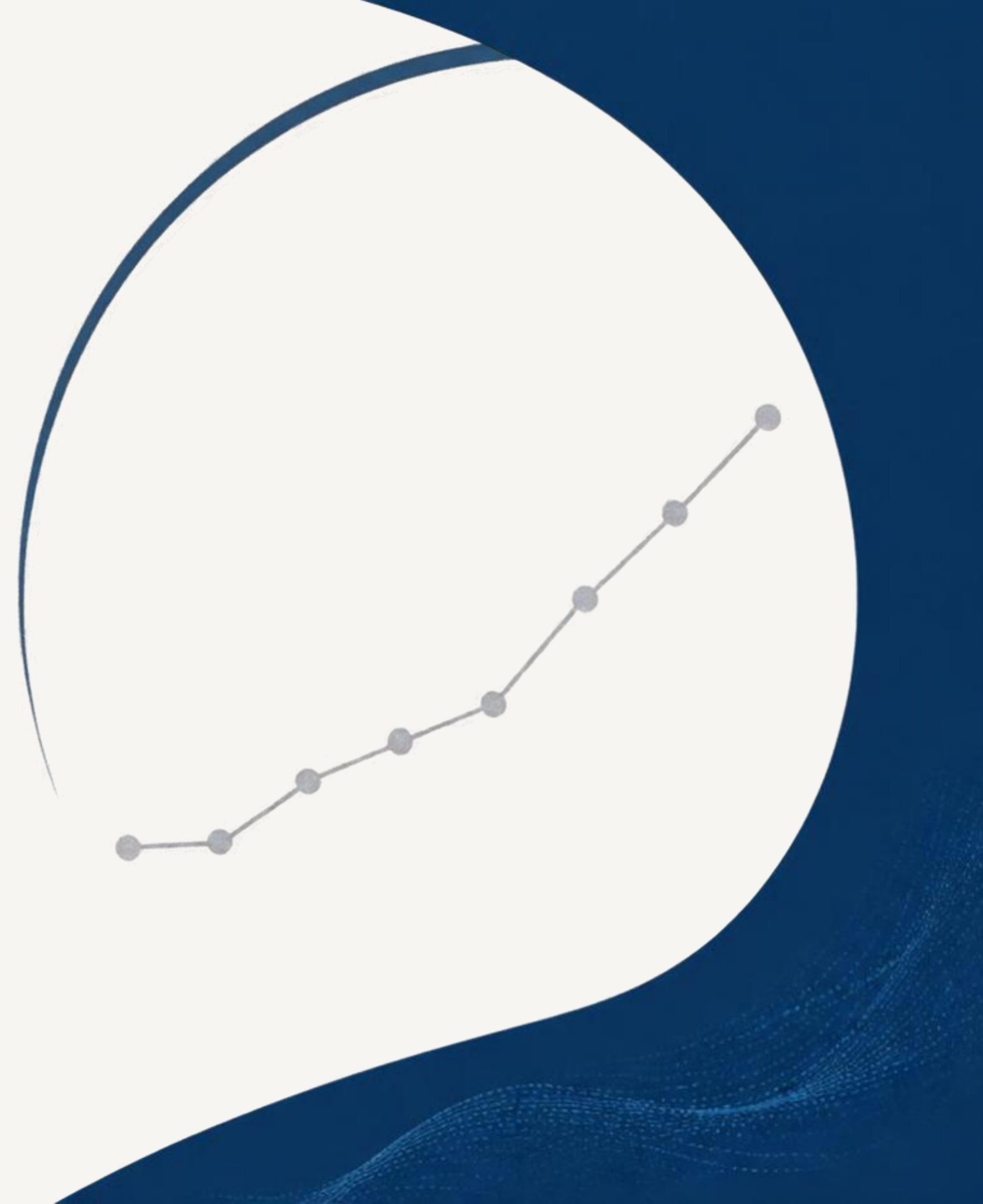


WALID NOFAL

Marketing Analysis Using Tableau

04 / 06

From market performance to
actionable business insight



Marketing Analysis Overview



Marketing analysis focuses on how demand is acquired, where it originates, and how efficiently each channel converts to confirmed, revenue-generating stays.

01

Customer Acquisition

02

Market Segments

03

Distribution Channels

04

Booking Volume & Revenue

05

Customer Origin

06

Lead Time & Repeat Guests

TOOLING

Tableau

Chosen for interactive filtering, blended data sources, and complex slicing across segment, channel, and geography.

Marketing KPIs & Dimensions



01	Total Bookings	Volume of reservations across the analytic window.
02	Revenue	Realized revenue: nights × ADR on confirmed bookings.
03	Average ADR	Average Daily Rate — pricing strength per stay.
04	Booking Share	Segment's proportion of total booking volume.
05	Cancellation Rate	Canceled ÷ total, measuring channel risk.
06	Average Lead Time	Days between booking and arrival.
07	Repeat Guests	Guests with a prior stay — loyalty signal.

Dashboard Walkthrough



- 1

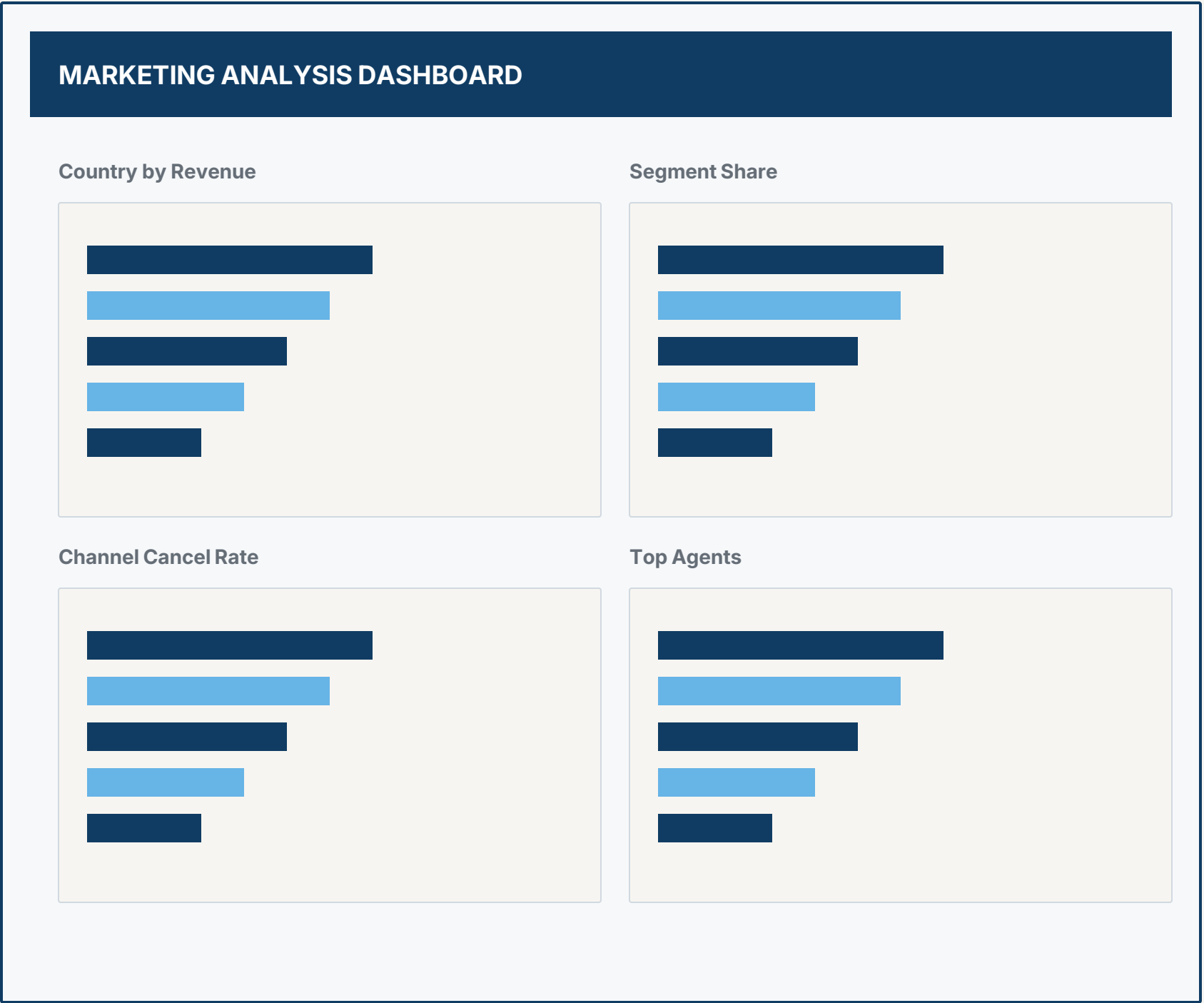
Top Country
Revenue and booking counts for top 1 country
- 2

Market Segment
Total Booking share, ADR, and revenue by segment.
- 3

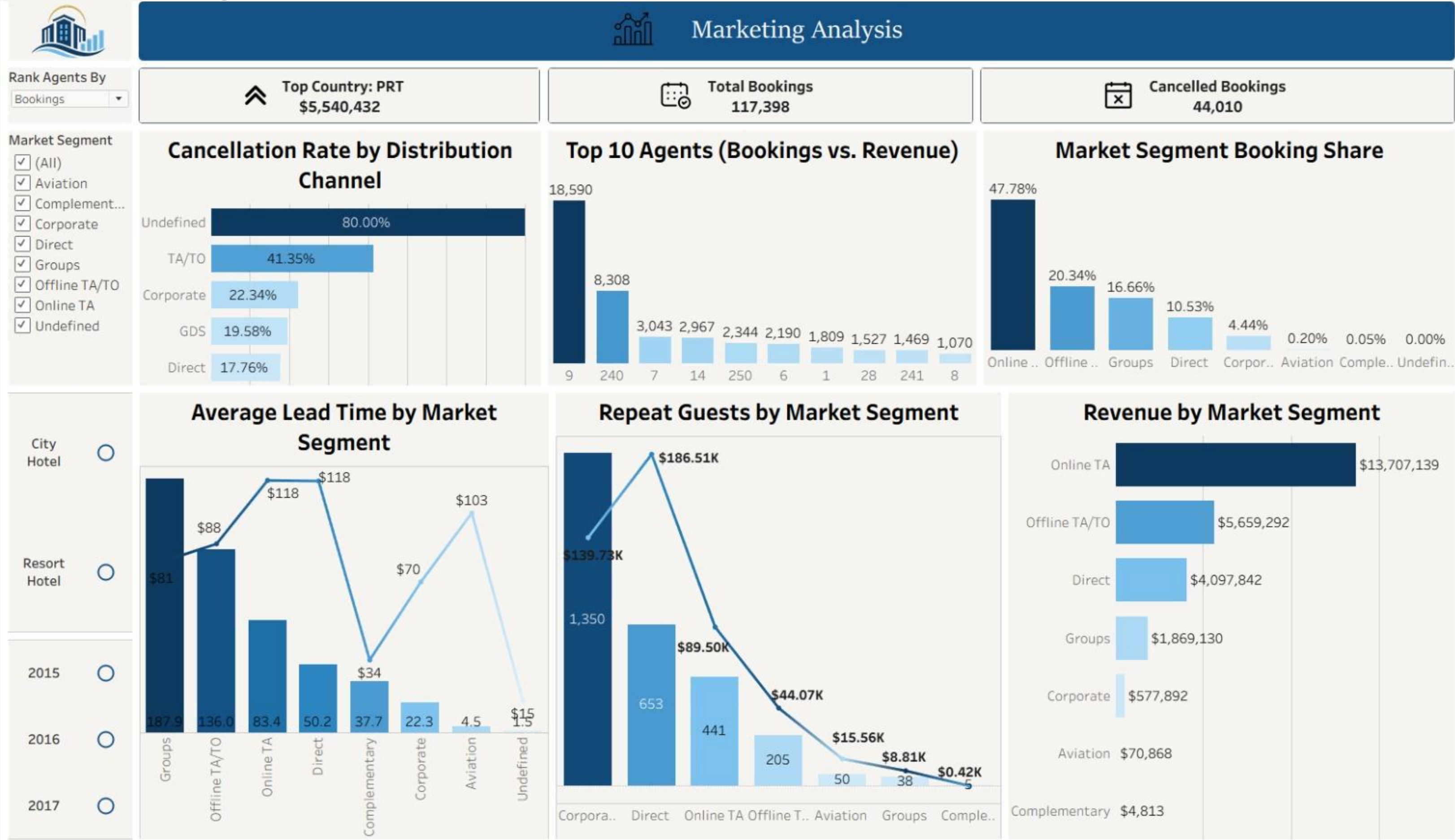
Distribution Channels
Cancellation rate and volume by channel.
- 4

Agent Performance
Top agents by bookings and revenue.
- 5

Filters & Slicers
Date range, hotel, and segment interactivity.



Market Segment Analysis



Marketing Dashboard — Findings

Portugal Top country by revenue	Online TA · 47.78% Largest booking share	TA/TO Highest cancellation rate channel
Agent 9 Top agent by bookings & revenue	Groups · 186.7d Longest avg lead time	Corporate Highest repeat-guest concentration

The Tableau dashboard surfaces market segment booking share, channel cancellation risk, agent performance rankings, average lead time by segment, and repeat-guest concentration — all filterable by date, hotel, and segment.

Marketing Insights



OBSERVED

Portugal is the top revenue country (~\$5.54M).

INTERPRETATION

Home market dominates; secondary EU markets are dispersed.

OBSERVED

Online TA holds ~47.78% booking share.

INTERPRETATION

OTA reliance concentrates channel risk; direct share is smaller.

OBSERVED

Lead time varies from ~5 days (Aviation) to ~187 days (Groups).

INTERPRETATION

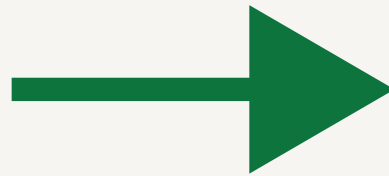
Segments differ in planning horizon; pricing windows must adapt.

OBSERVED

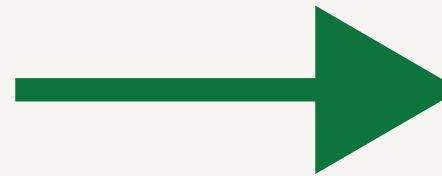
Cancellation rates diverge by distribution channel.

INTERPRETATION

Channel-specific policies (deposits, cutoffs) should be re-evaluated.

[illegible]

	A ^B C arrival_date_year		ABC month		A ^B C arrival_date_month		A ^B C arrival_date_day_of_month	
	Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	
	2015		7		July		1	24
	2015		7		July		1	

[illegible][illegible]

HANA MOHAMED

SQL Query 3 • Cancellation Analysis + Power BI

05 / 06

From cancellation patterns to operational insight



Cancellation Analysis — Overview

1 Cancellation rate by lead time

2 Cancellation by deposit type

3 By room type changes

4 By arrival month

5 Special requests vs cancellation

6 Previous cancellation history

7 Booking changes vs cancellation

8 Total bookings

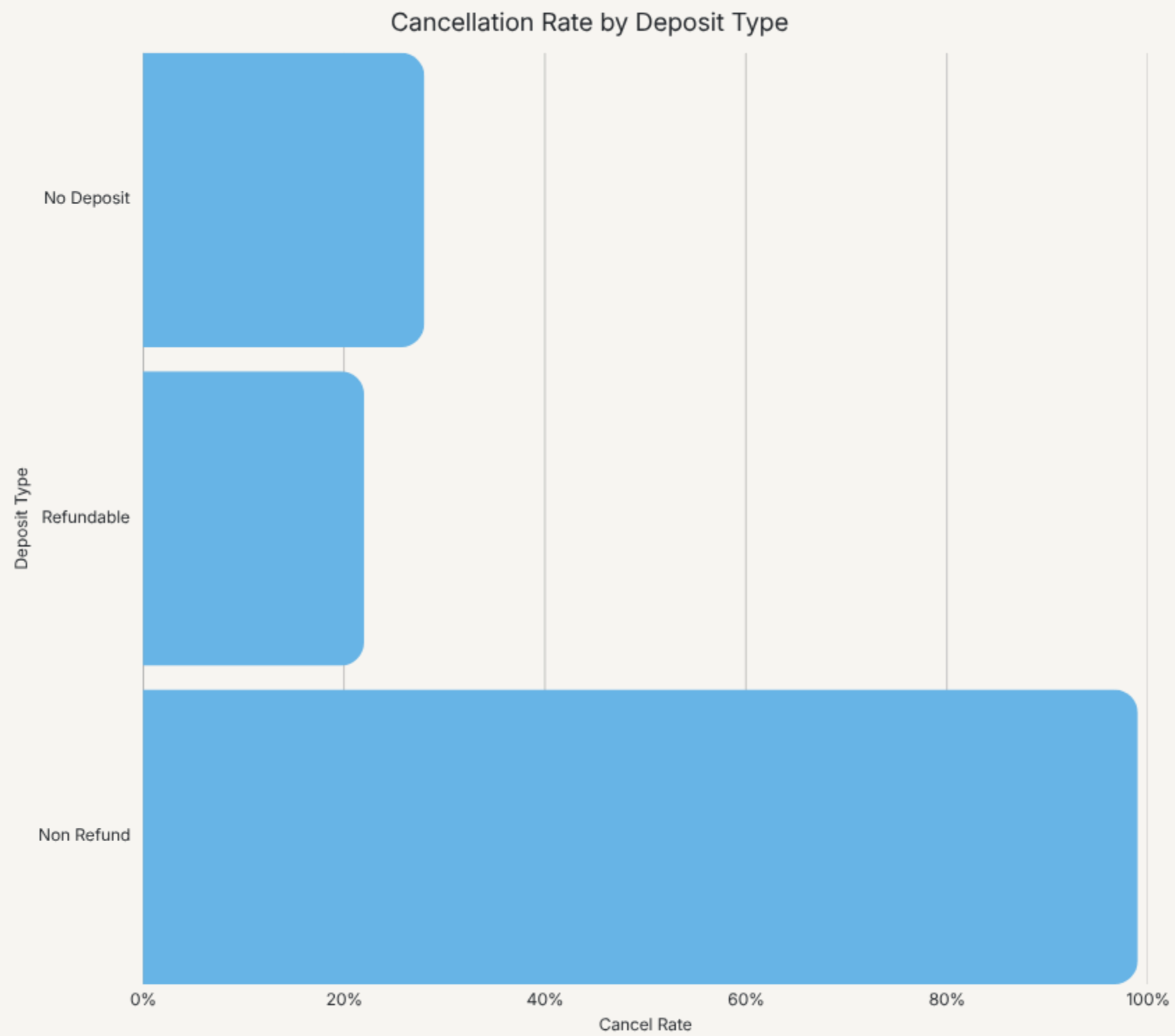
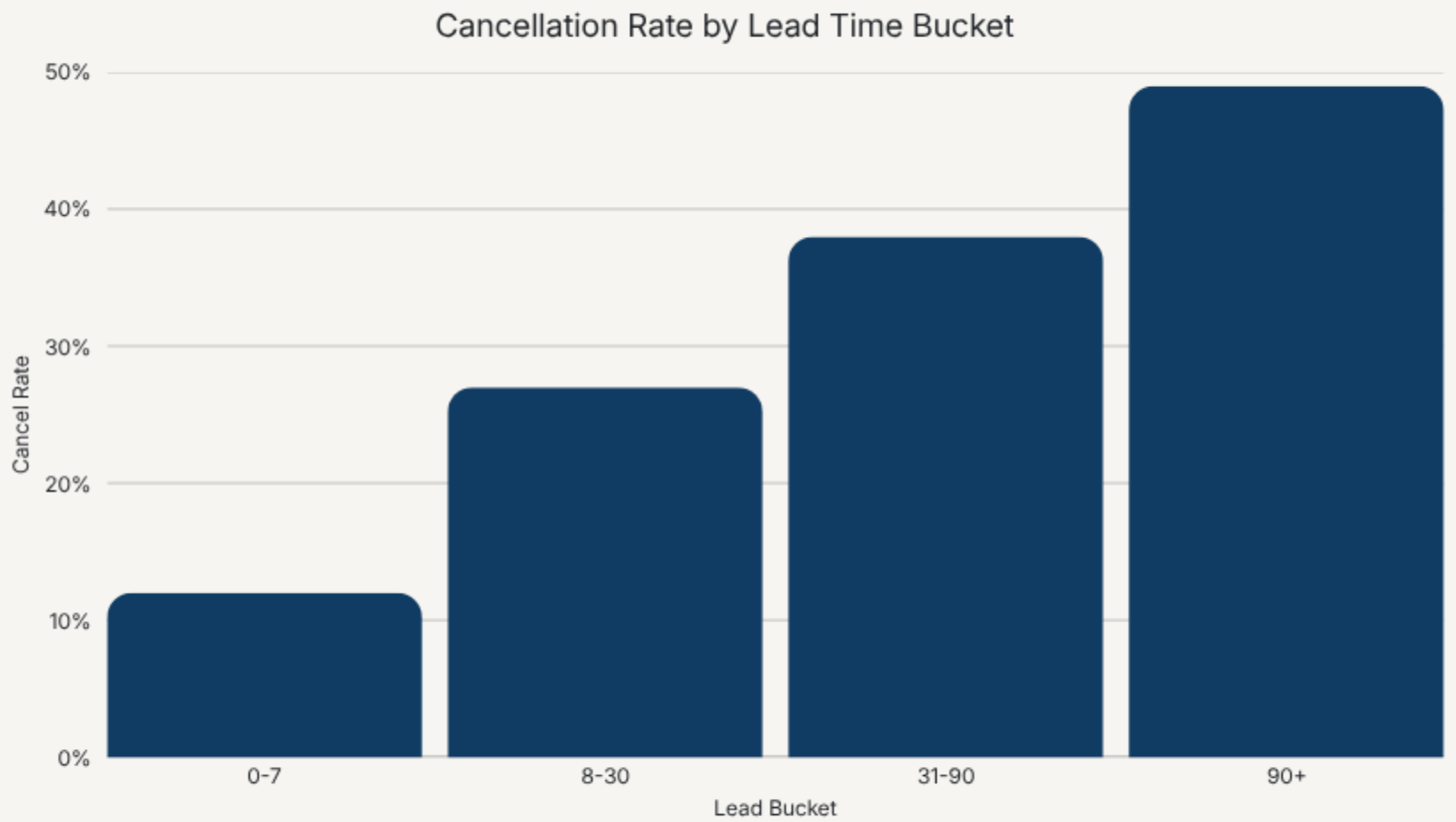
9 Overall cancellation rate

WHY IT MATTERS

Cancellations sit at 37.5% and translate to \$16.72M in lost revenue — the single largest lever for revenue recovery and forecasting accuracy.

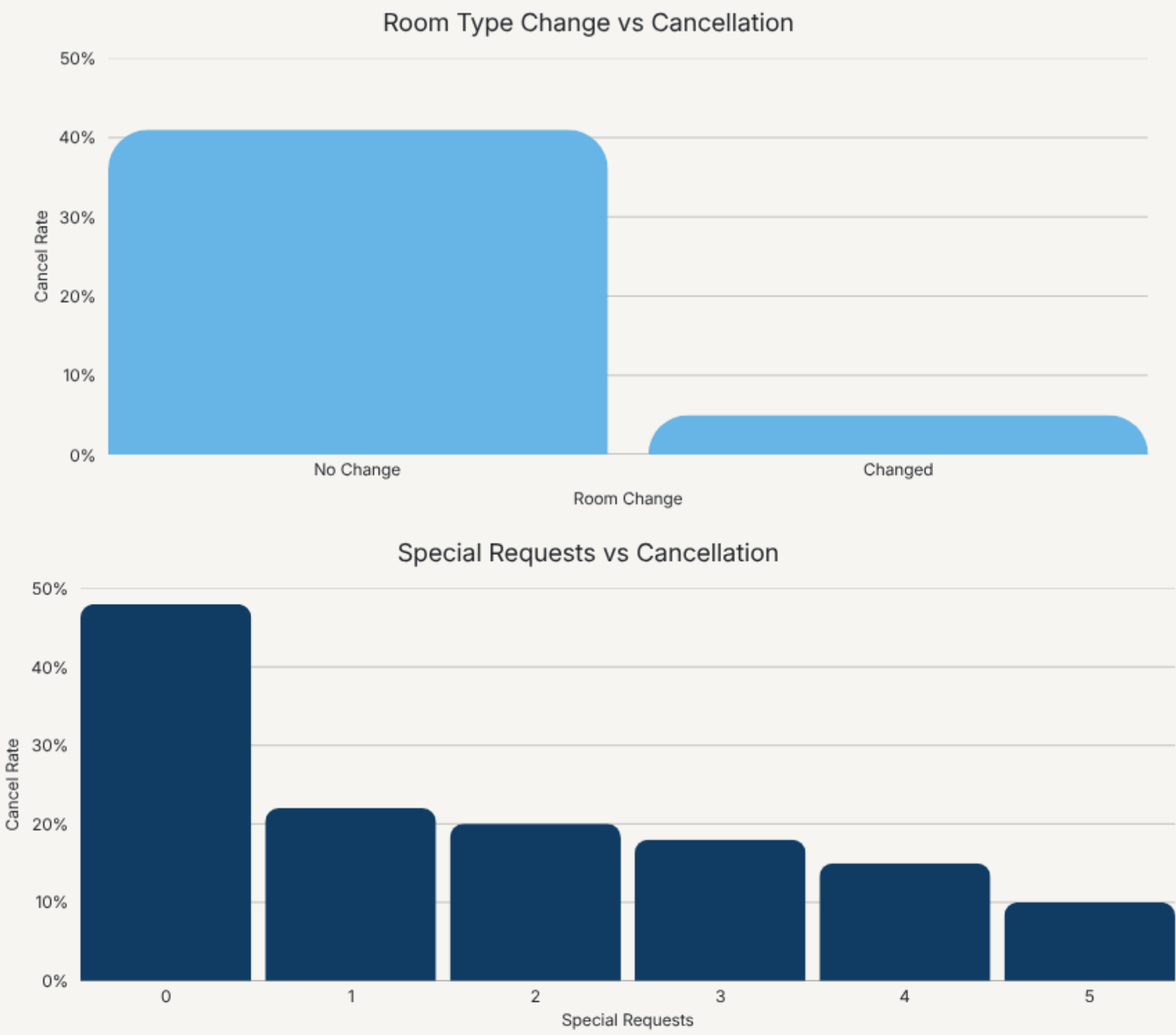
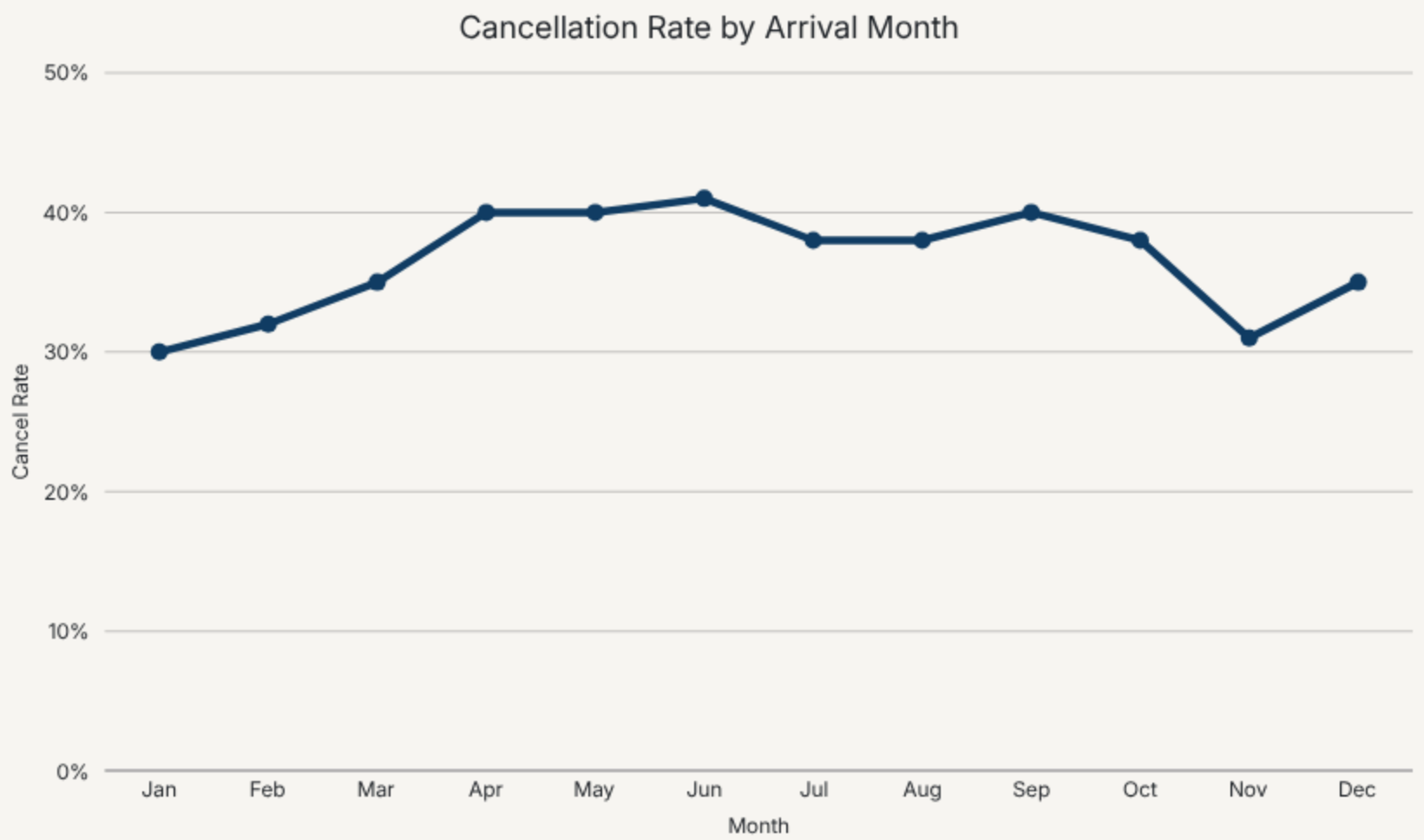
Cancellation by Lead Time & Deposit

```
SELECT
  CASE WHEN lead_time<=7 THEN '0-7'
        WHEN lead_time<=30 THEN '8-30'
        WHEN lead_time<=90 THEN '31-90'
        ELSE '90+' END AS lead_bucket,
  AVG(CAST(is_canceled AS FLOAT))*100
  AS cancel_rate_pct
GROUP BY 1;
```



Room Change · Arrival Month · Special Requests

```
SELECT arrival_date_month,
       SUM(CAST(is_canceled AS INT)) AS canceled,
       AVG(CAST(is_canceled AS FLOAT))*100 AS rate_pct
FROM hotel_bookings_cleaned
GROUP BY arrival_date_month;
```



Previous Cancellations & Booking Changes

117,398

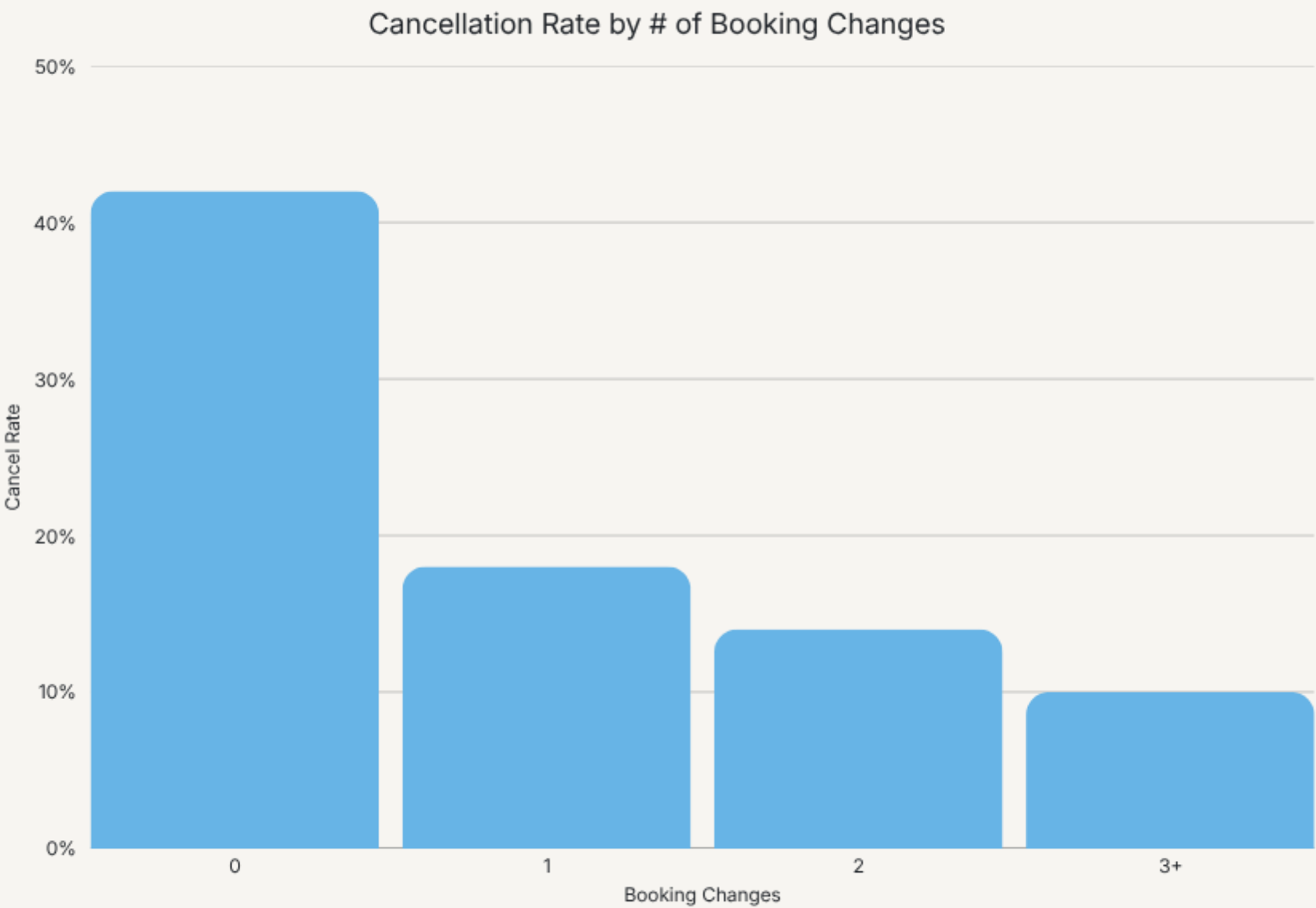
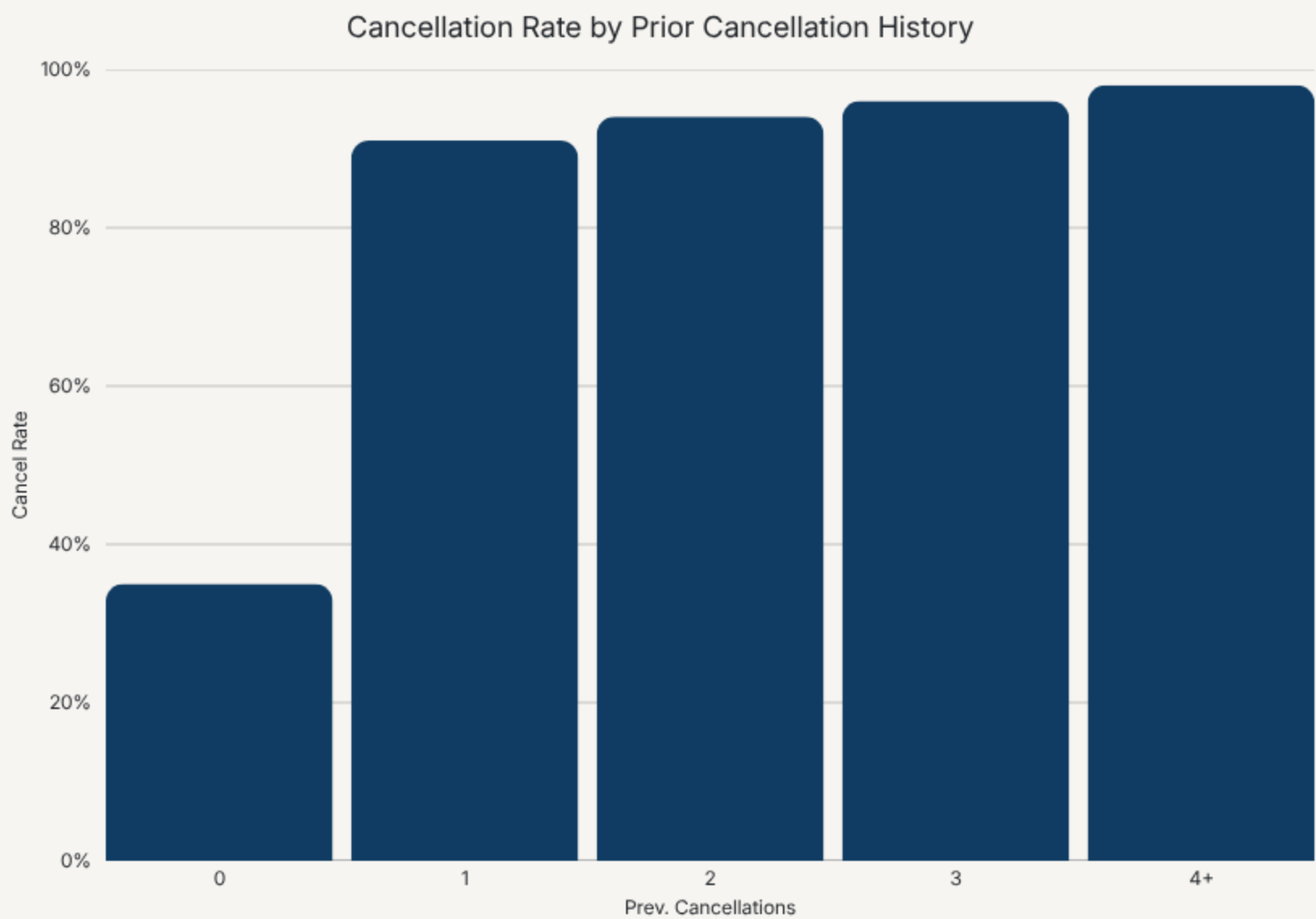
TOTAL BOOKINGS

44,020

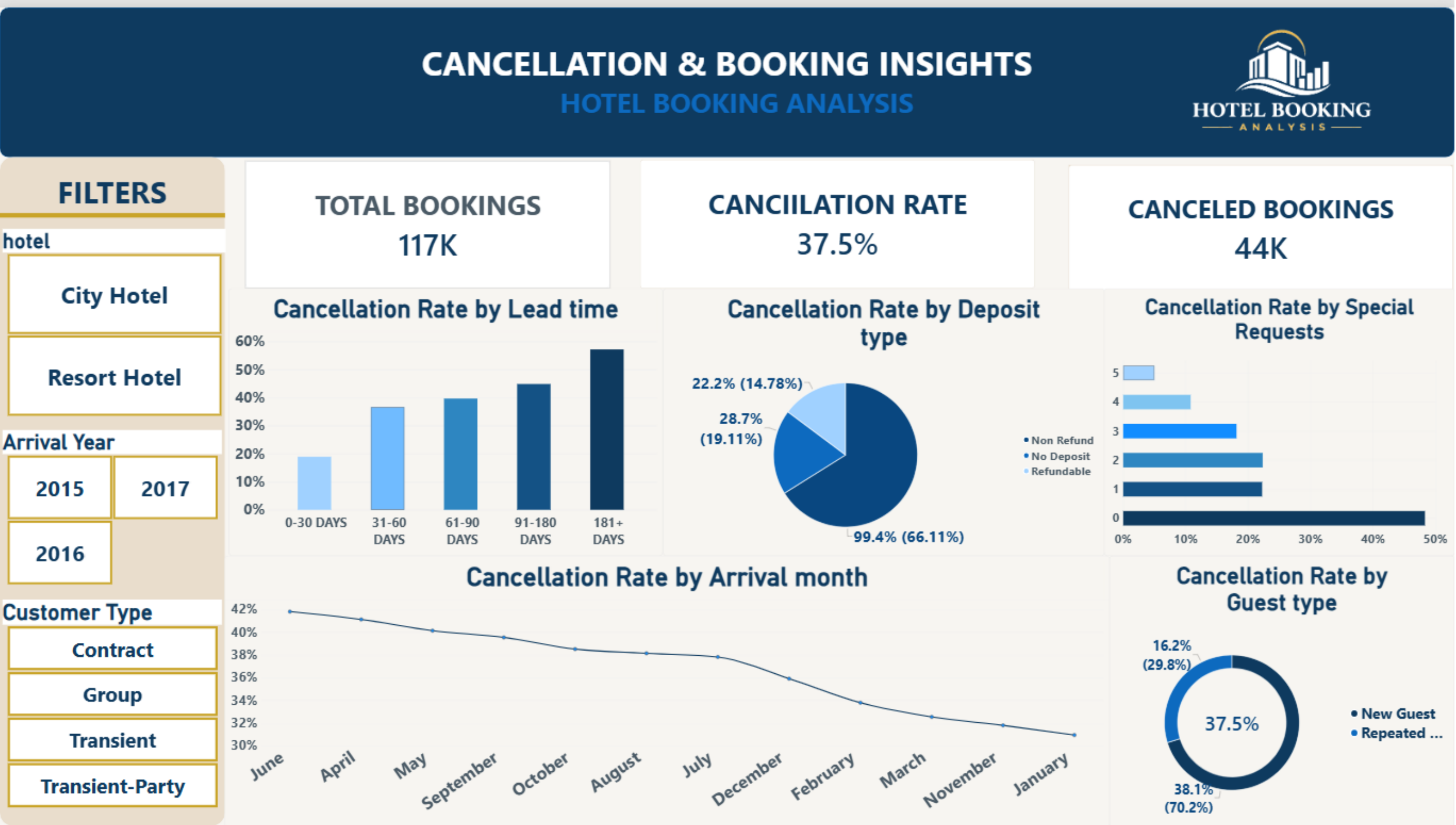
CANCELLATIONS

37.5%

OVERALL CANCEL RATE



Cancellation & Booking Dashboard



Cancellation Insights



OBSERVED PATTERN

Cancellation rate rises steadily with lead time.

POSSIBLE INTERPRETATION

Longer horizons mean more plan changes; deposits or reconfirmation windows can moderate risk.

OBSERVED PATTERN

Non-refundable deposits show near-universal cancellation.

POSSIBLE INTERPRETATION

Suggests these bookings may represent speculative or duplicate reservations rather than active demand.

OBSERVED PATTERN

New guests cancel more often than repeat guests.

POSSIBLE INTERPRETATION

Loyalty is a strong retention signal; repeat-guest incentives can protect volume.

OBSERVED PATTERN

Zero special requests correlate with higher cancellations.

POSSIBLE INTERPRETATION

Engagement (requests, changes) appears associated with commitment to arrival.

EMAN ELNAHAS

SQL Query 4 • Customer Behavior + Power BI Dashboard



06 / 06

From guest behavior to
the final project takeaways



Customer Behavior — Overview

Query 4 profiles who books, what they want, and how they interact with the hotel — the operational and marketing complement to revenue and cancellation analytics.

01

Guest Demographics

02

Preferences & Amenities

03

Service Intensity

04

Booking Changes

05

Room Discrepancies

06

Loyalty & Repeat

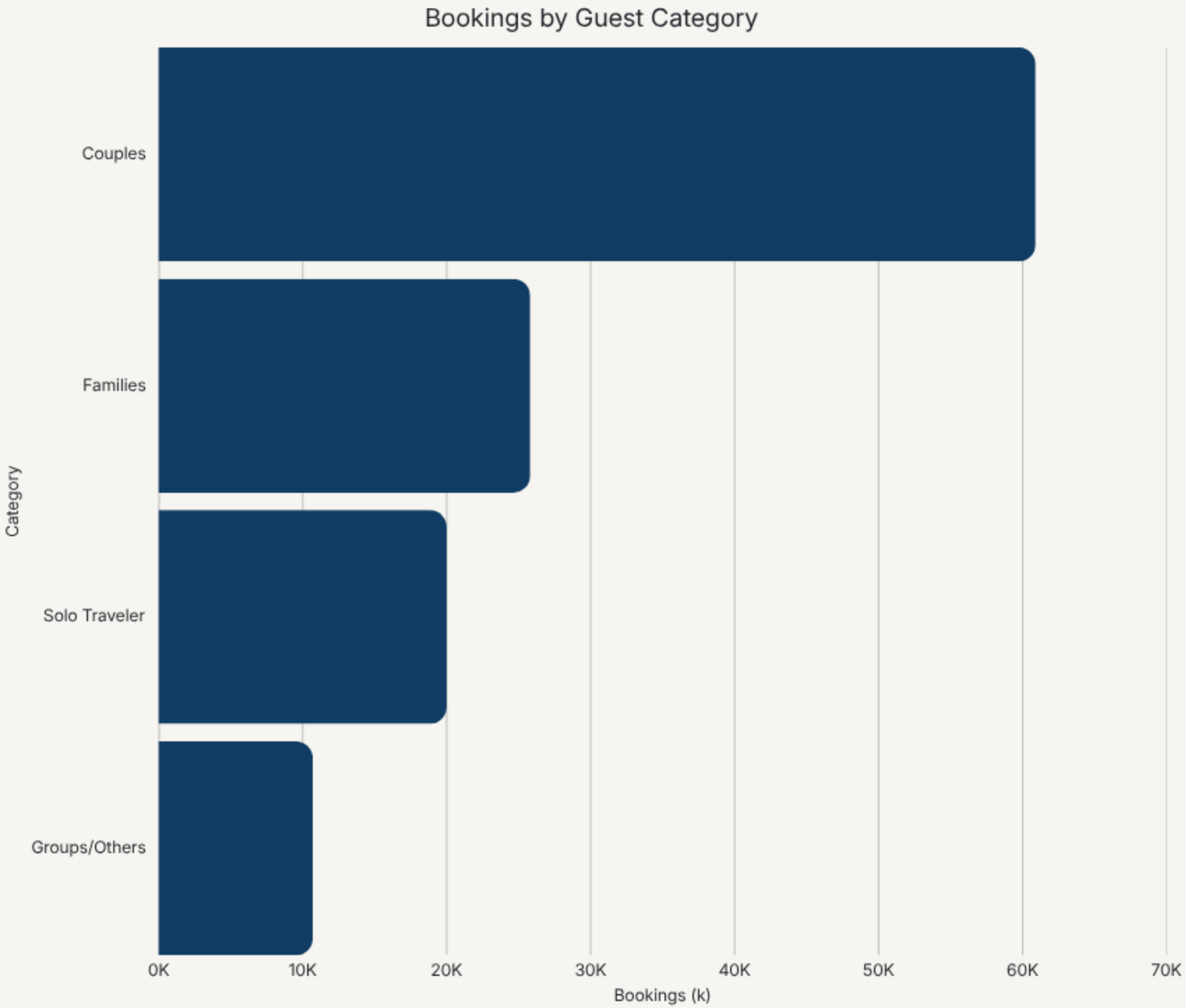
Findings inform staffing plans, amenity investment, loyalty programs, and targeted marketing campaigns.

Guest Demographics



```
SELECT
CASE
  WHEN adults=1 AND children=0 AND babies=0
    THEN 'Solo Traveler'
  WHEN adults=2 AND children=0 AND babies=0
    THEN 'Couples'
  WHEN children>0 OR babies>0 THEN 'Families'
  ELSE 'Groups/Others'
END AS guest_category, ...
GROUP BY guest_category;
```

Couples 52% 60,900 bookings Avg stay: 3.4 nights	Families 22% 25,800 bookings Avg stay: 4.1 nights
Solo 17% 20,000 bookings Avg stay: 2.8 nights	Groups/Others 9% 10,700 bookings Avg stay: 2.6 nights

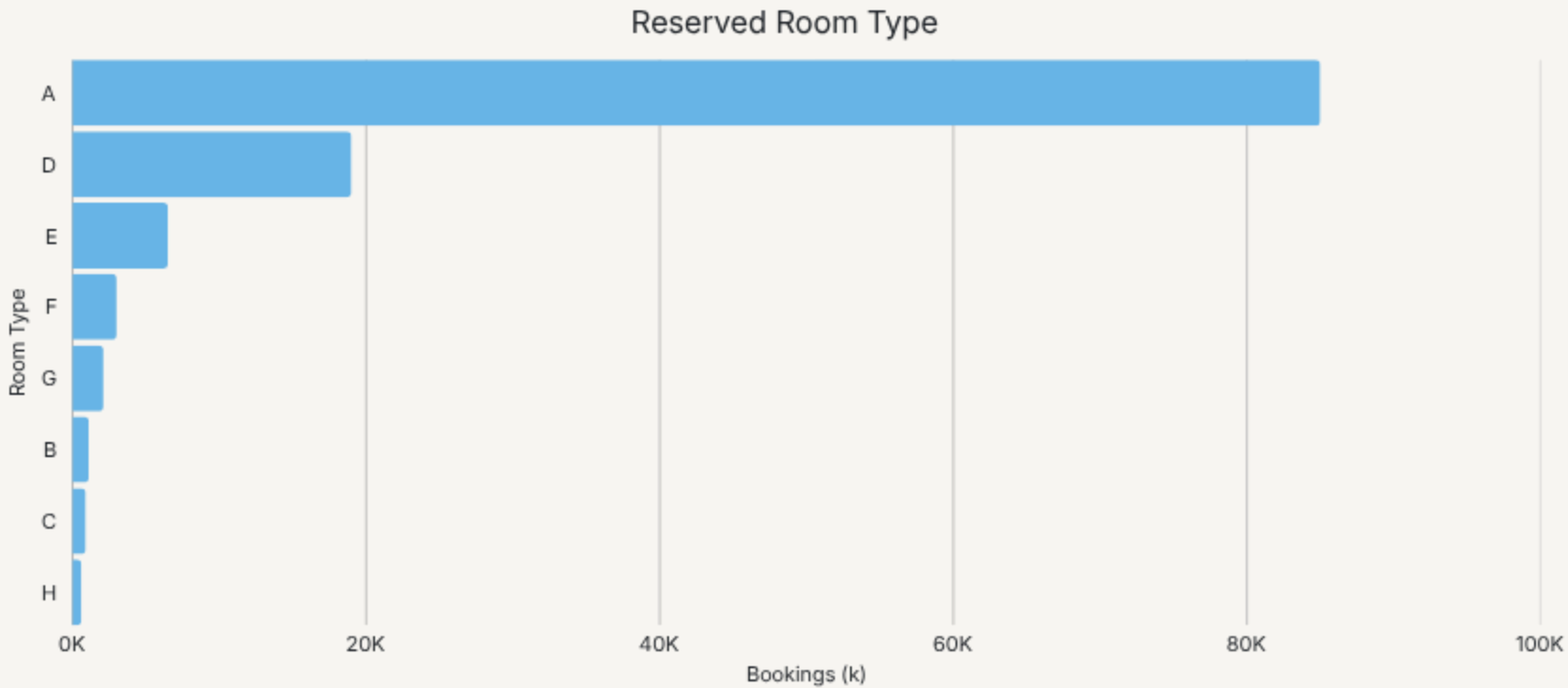
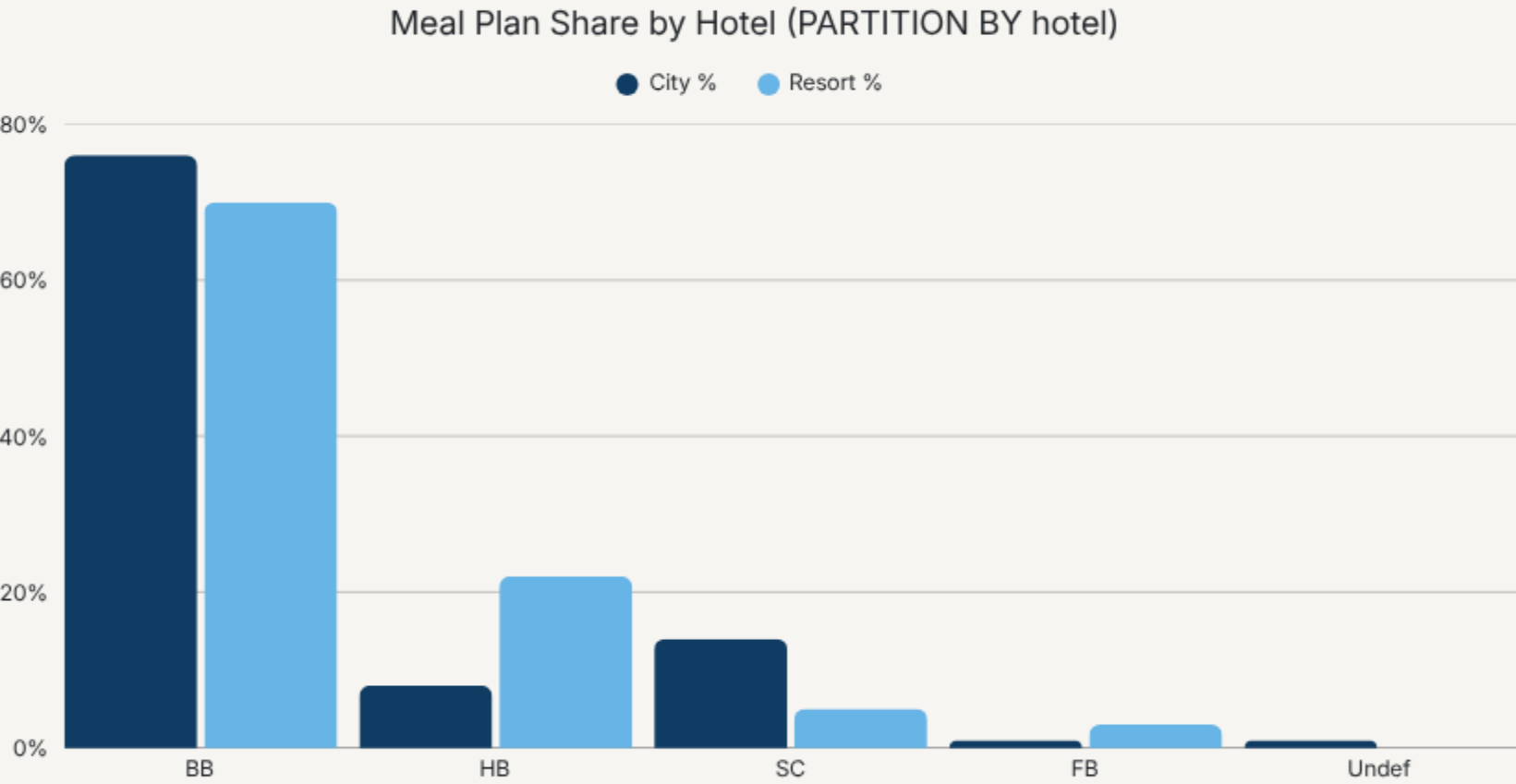


Guest Preferences & Amenities



```
SELECT hotel, meal,
       COUNT(*) AS bookings,
       100.0 * COUNT(*)
         / SUM(COUNT(*)) OVER (PARTITION BY hotel)
         AS meal_share_pct
GROUP BY hotel, meal;
```

PARTITION BY hotel isolates each property's mix — so a City breakfast share is not diluted by Resort half-board volume.



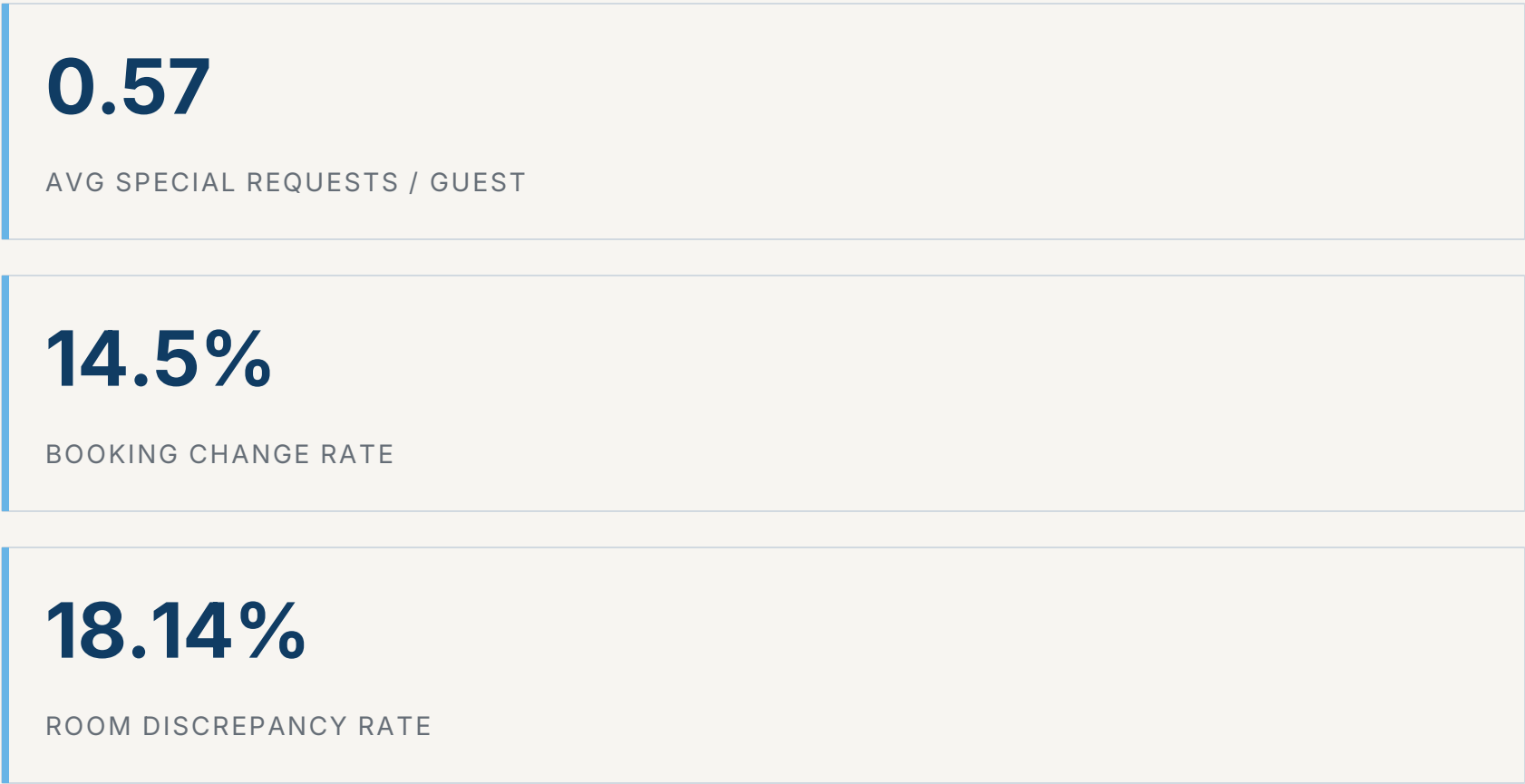
Service Intensity & Operations



```
SELECT hotel,
  AVG(total_of_special_requests) AS avg_requests,
  100.0 * SUM(CASE WHEN booking_changes>0
    THEN 1 ELSE 0 END) / COUNT(*)
    AS change_rate_pct,
  100.0 * SUM(CASE WHEN reserved_room_type
    <> assigned_room_type
    THEN 1 ELSE 0 END) / COUNT(*)
    AS room_discrepancy_rate_pct
GROUP BY hotel;
```

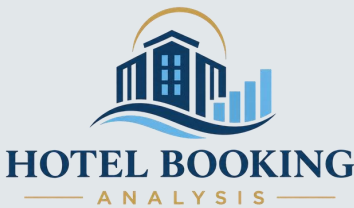
Room discrepancy — reserved type ≠ assigned type — is a leading indicator of on-property operational friction and guest satisfaction risk.

Booking-change rate captures how often guests modify reservations before arrival, a proxy for engagement and reduced cancellation risk.



Customer Behavior Dashboard





COMPLETE PROJECT PIPELINE

From raw records to business insight.



REVENUE \$42.7M realized · \$16.7M lost	MARKETING OTA-heavy · Portugal top origin	CANCELLATION 37.5% overall · lead time driven	BEHAVIOR 18% room discrepancy · low repeat
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Thank You